

3D Architecture

01



Both the buildings behind me are useful. They have a unique design and are planned in a specific manner. Let's plan such type of 3D architecture.



Buildings are all around us. Each one is constructed to fulfil certain requirements. The art or practice of designing and constructing buildings is called architecture. It requires construction skills as well as an aesthetic sense. Some are for residential use, while others are for commercial purposes or religious places. A few buildings are designed to be both beautiful and functional.

Architectural types



Different types of architecture are established by individuals or society. They are constructed to meet the requirements of different institutions. Constructing an aesthetic and purposeful structure requires planning.

An **Architectural plan** identifies the need for a construction, plans for the materials required, analyse available resources, and determines financial support. Architectural planning requires the use of specific software. We will work on an activity to create an ancient structure.

Let us create a 3D visualization of the part of the Al Jahili fort present in UAE.



Image credits: Ruler's Representative Court.

Al Jahili, located at the heart of the Garden City, stands as both one of the UAE's largest and oldest forts. Built in the early 19th century, its construction utilized mudbrick, straw, and palm tree fibres. It was originally designed to protect the nearby oasis and its plentiful date palms, and it holds historical significance to this day.

In the present day, this historic structure has undergone meticulous restoration and now serves as a museum. Within its walls, visitors can explore a diverse range of exhibitions, including a captivating black-and-white photography series captured by renowned British adventurer Sir Wilfred Thesiger. These compelling images were taken during his daring expedition across the Rub Al Khali (Empty Quarter) desert in the 1940s.

Activity

Create a 3D visualisation of a small portion of the Al Jahili Castle.

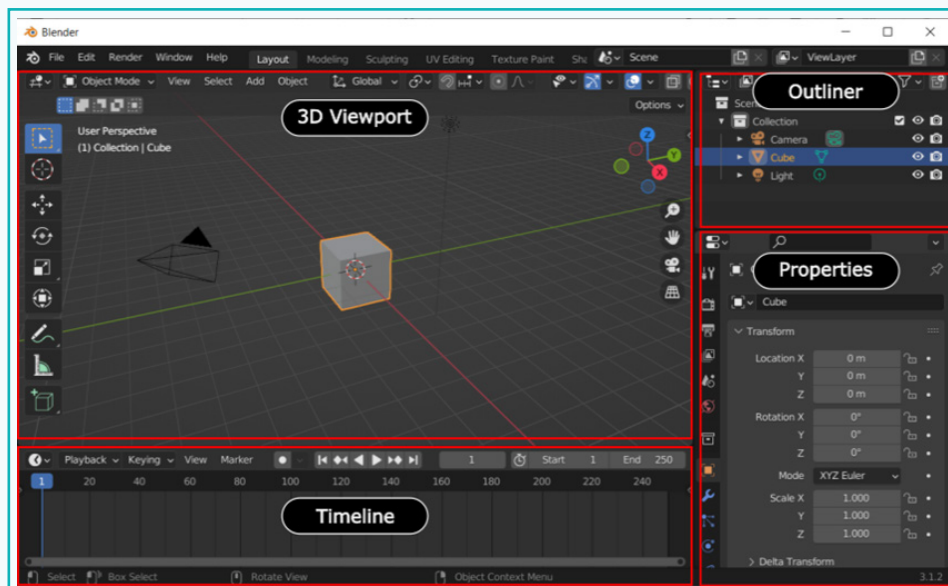


Image credits: Rachel's Ruminations

We will use a 3D application software to create a 3D visualisation of this architecture. We have several software options, out of which Blender is an open source and offers useful features.

Blender interface consists of several windows. Blender's default start-up layout consists of the following Editors:

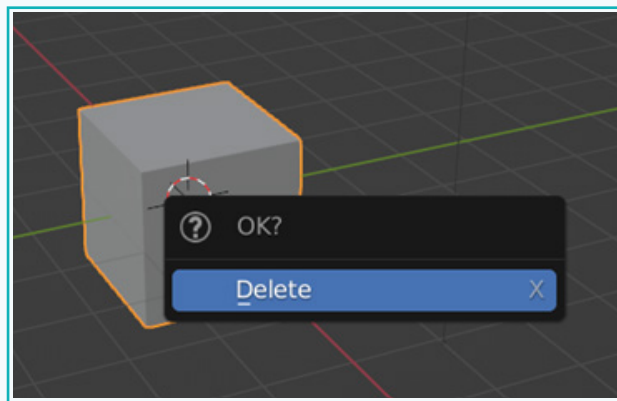
- 3D Viewport (top left)
- Outliner (top right)
- Properties (bottom right)
- Timeline (bottom left)



Activity

1 Delete the existing object.

- When we start the application, we have a cube, a camera, and a light in the viewport by default.
- Delete the cube and create a new object. To delete an object, select it and press 'Delete'.



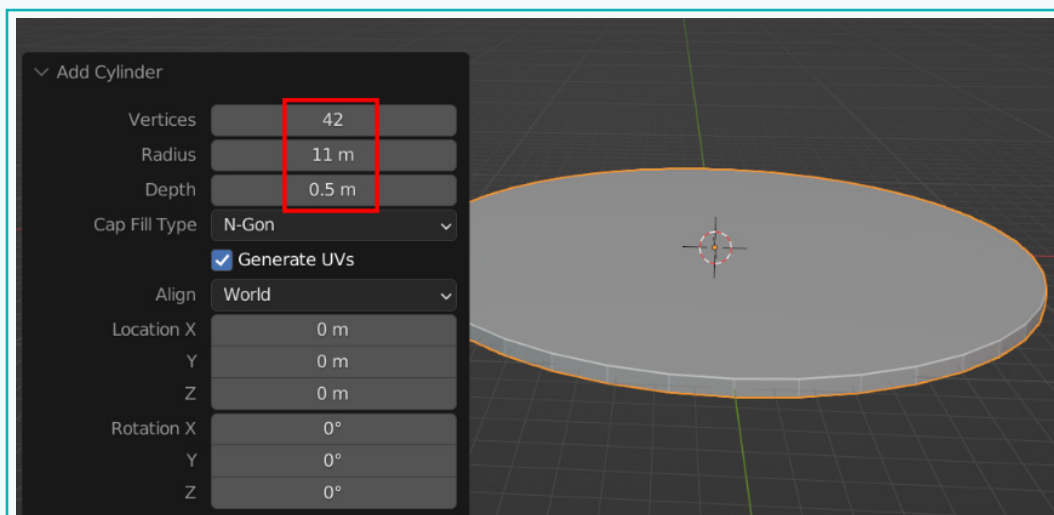
Now that we have an empty scene, let us try to add some elements to the scene.

The Al Jahili Castle has a circular base structure comprising of three cylindrical shapes of decreasing diameters stacked one on top of the other.

Create the Base of the Castle

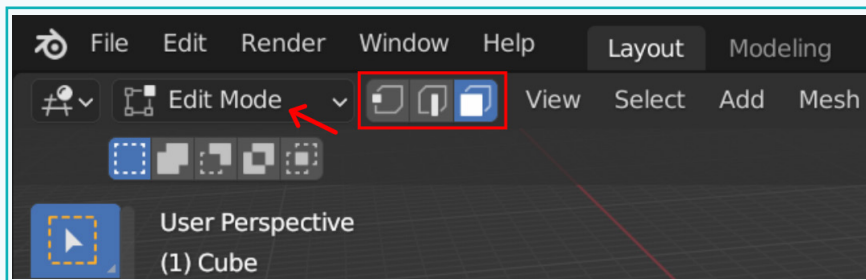
Let us start by creating the base of the Al Jahili castle.

2 Press 'Shift + A' → 'UV Cylinder'. Specify the dimensions, as shown in the image.



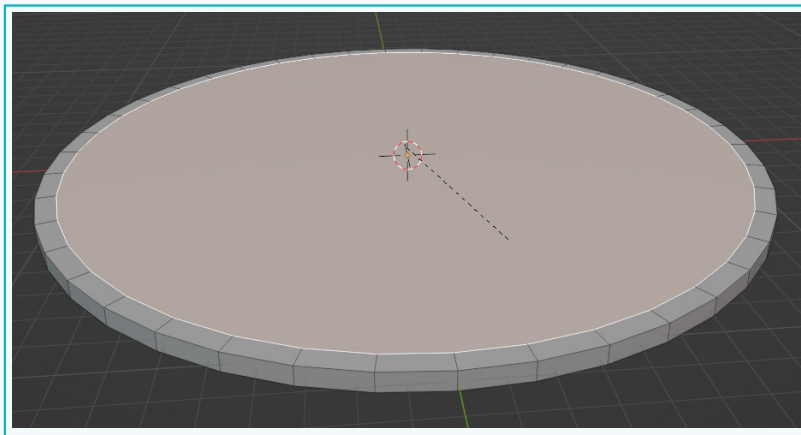
Activity

- 3 Press 'Tab' to go into the 'Edit' mode and select the 'Face Selection' mode. Refer the image below to switch between the three select modes.

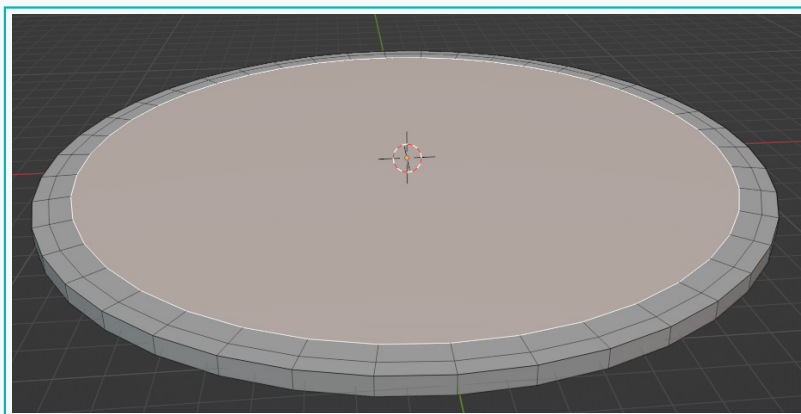


- 4 Select the top face and press 'I' to create an inset of the selected face.

The Inset tool creates a new face on the surface of the selected face.

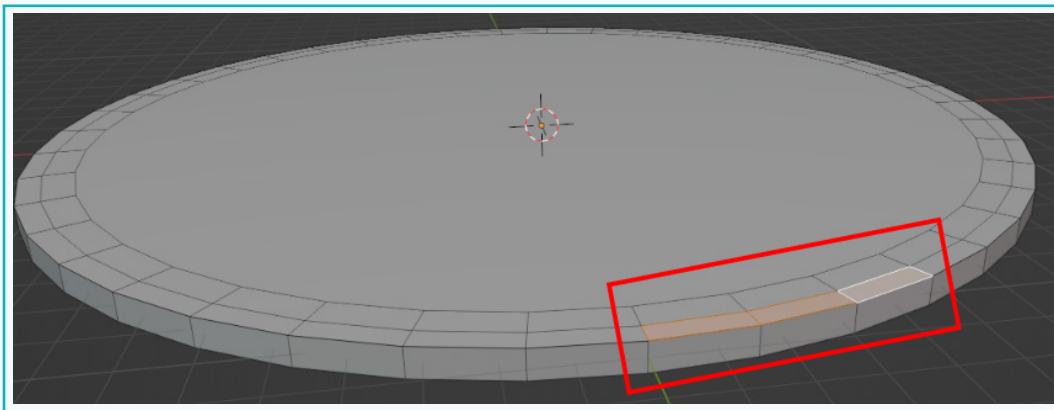


- 5 Create another inset on top of the previous one.

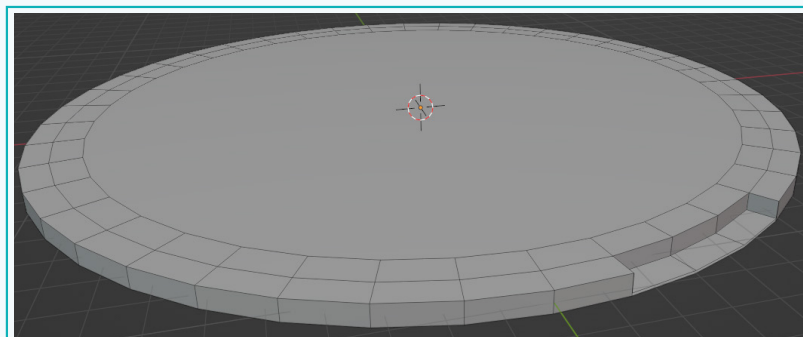
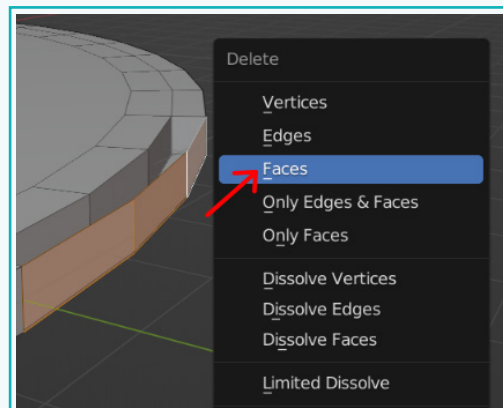
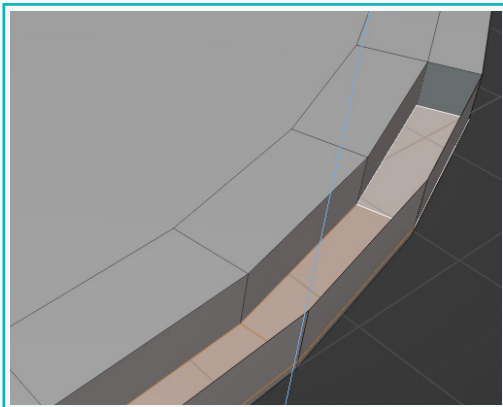


Activity

- 6 Next, make sure we are in the 'Face Selection' mode by clicking and selecting three consecutive faces, as shown.



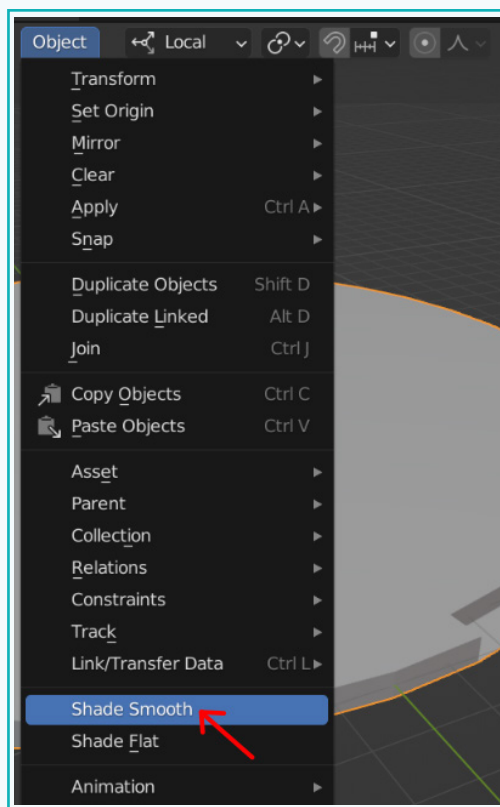
- 7 Press 'E' to extrude the selected faces downwards. Select the surrounding faces on the side and press 'X' to remove them.



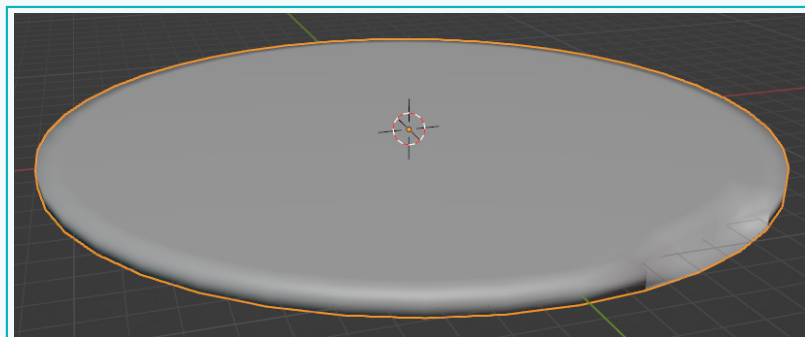
Activity

The '**Extrude tool**' is an edit mode tool that allows you to extend a selected portion of an object in any direction.

- 8 Let us smoothen the geometry by selecting the 'Object' menu → 'Shade Smooth'.

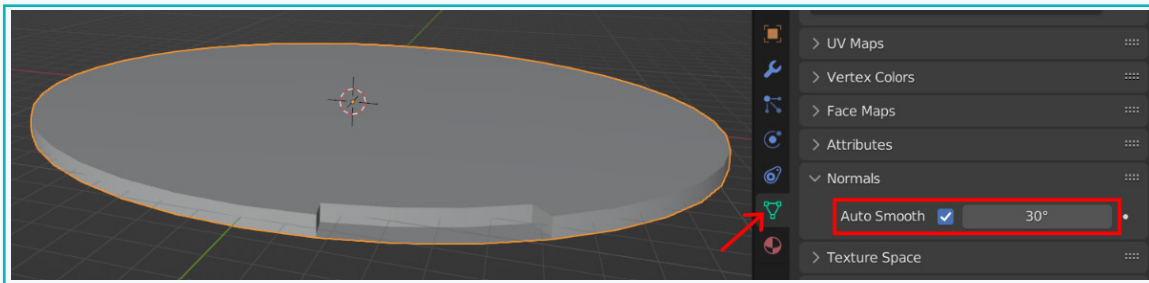


Sometimes, the 'Shade Smooth' function may not work accurately, resulting in outputs as shown below.



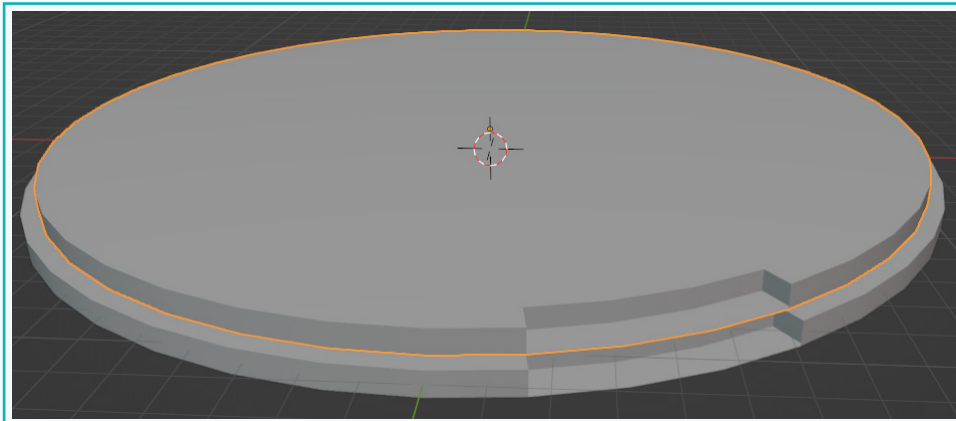
Activity

- 9 This can be rectified by clicking on the 'Object Data Properties' window and under the 'Normals' property, enabling the 'Auto Smooth' option.



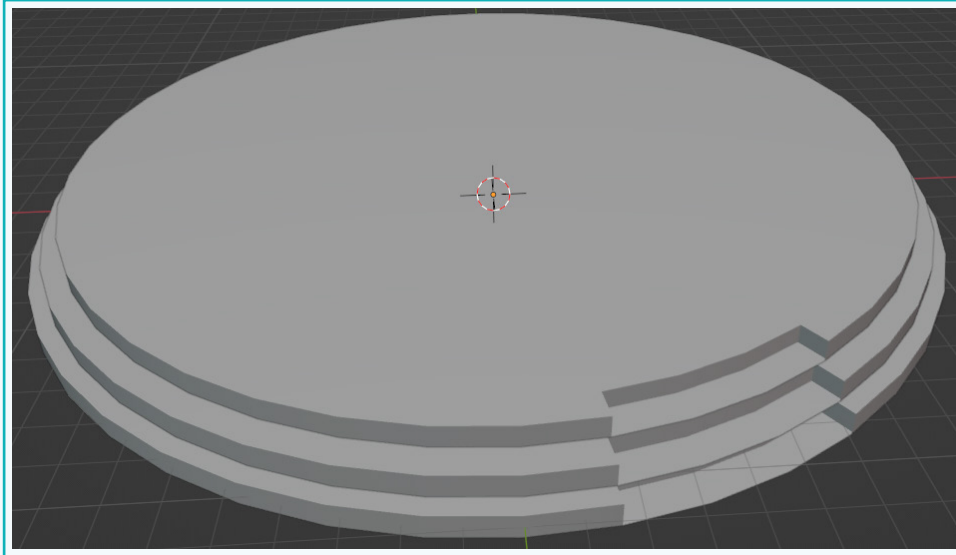
Now that we have one cylindrical shape ready, we can just duplicate it to create the remaining two cylindrical shapes for the base.

- 10 Press 'Shift + D' followed by 'Z' to move the duplicated shape upwards on the z-axis. Left click to confirm, then press 'S' to scale the shape down.

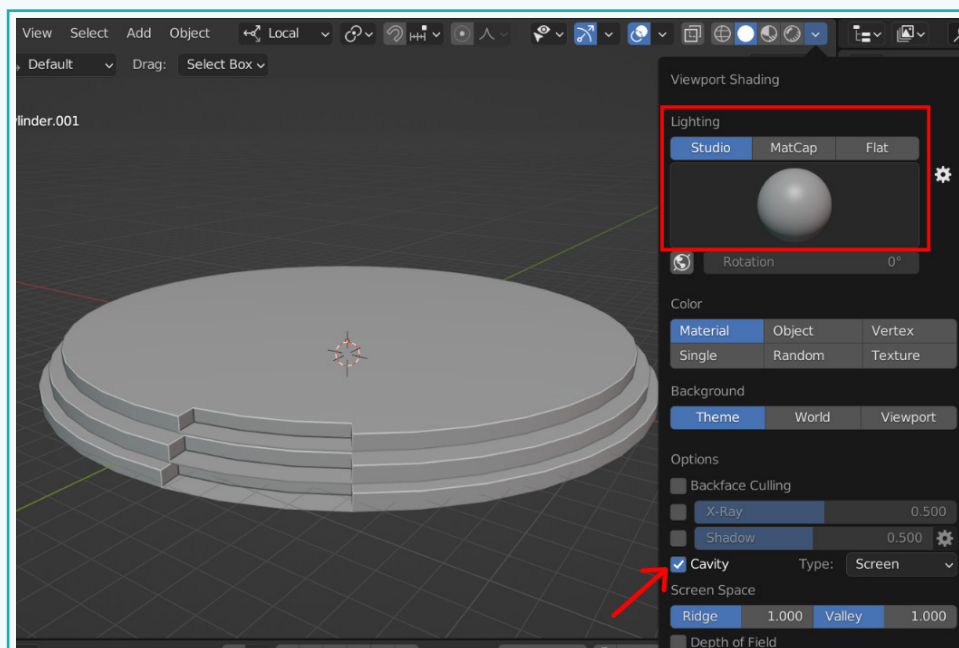


Activity

- 11 Similarly, create the topmost cylindrical shape.



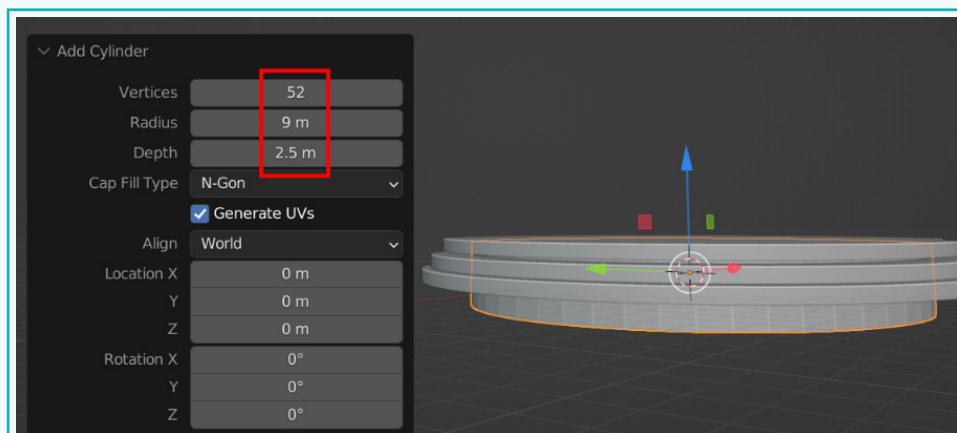
- 12 To view the model more clearly, enable the 'Cavity' option under the 'Solid Shading' mode. Also, experiment with the 'Lighting' options.



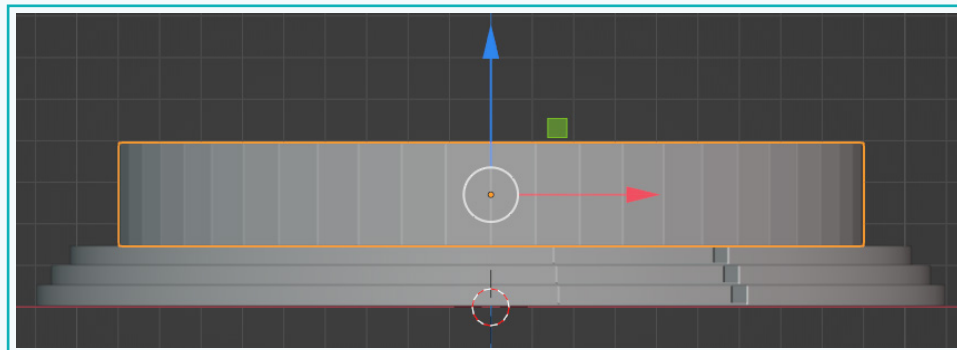
Activity

Let us continue with the next structure on top of the cylindrical base, which is also a cylinder.

- 13 Create a cylinder with the specified dimensions.

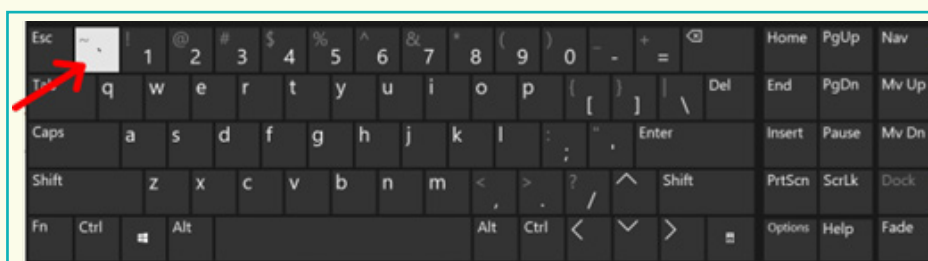


- 14 Press '1' to go into the front view. Move the cylinder upwards to place it on top of the base structure.



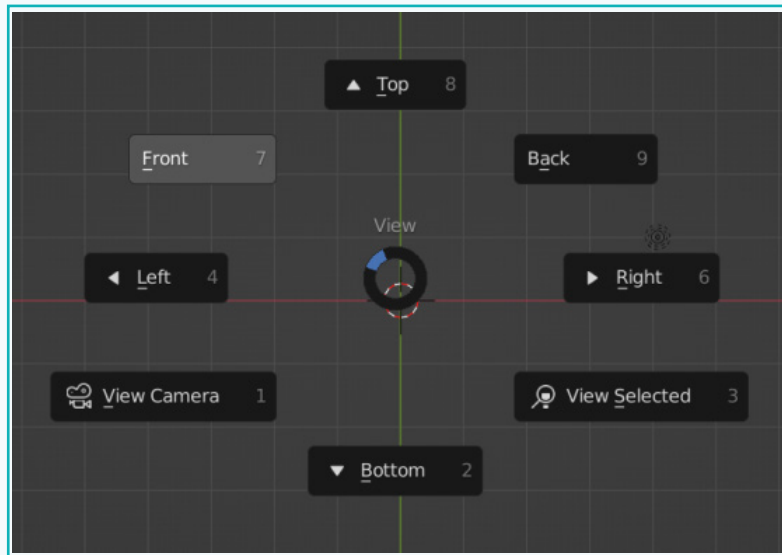
Note:

If the Numpad is not present, press the 'Backtick (`)' button on the keyboard.

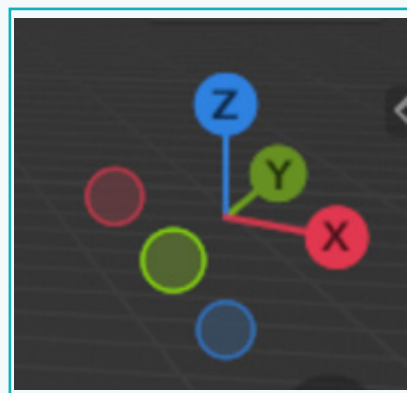
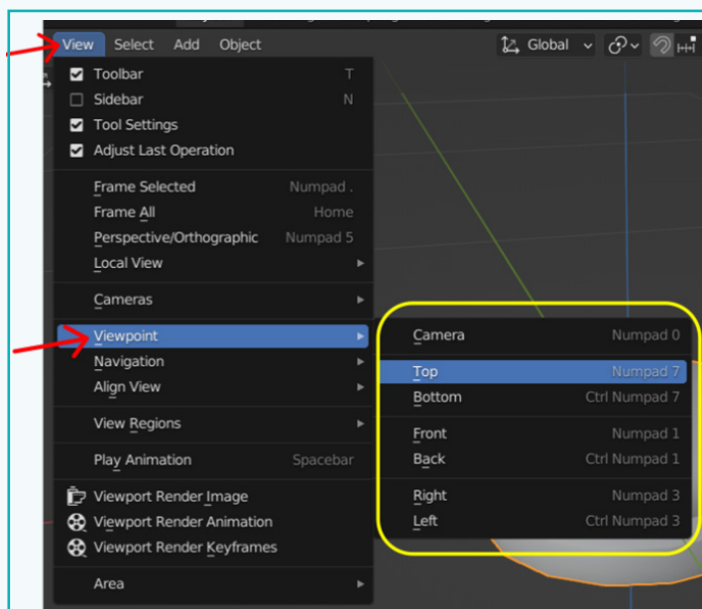


Activity

This will open a 'View' pie menu, allowing you to effortlessly swap between different views.

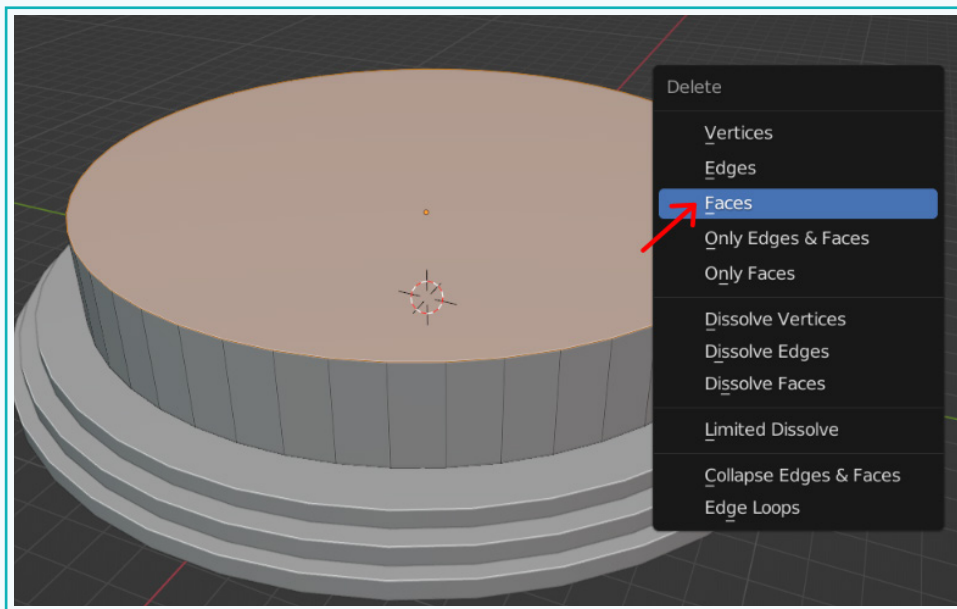


These options can also be accessed in the 'View' → 'Viewpoint' menu.

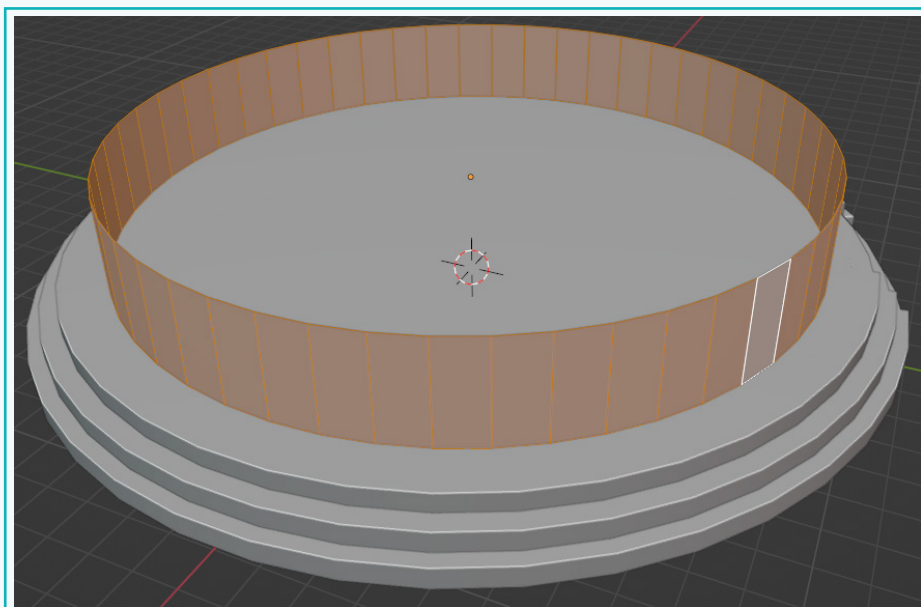


Activity

- 15 Press 'Tab' to go into the 'Edit' mode. Select the 'Face Selection' mode. Select the topmost face and press 'X' to delete the selected face.

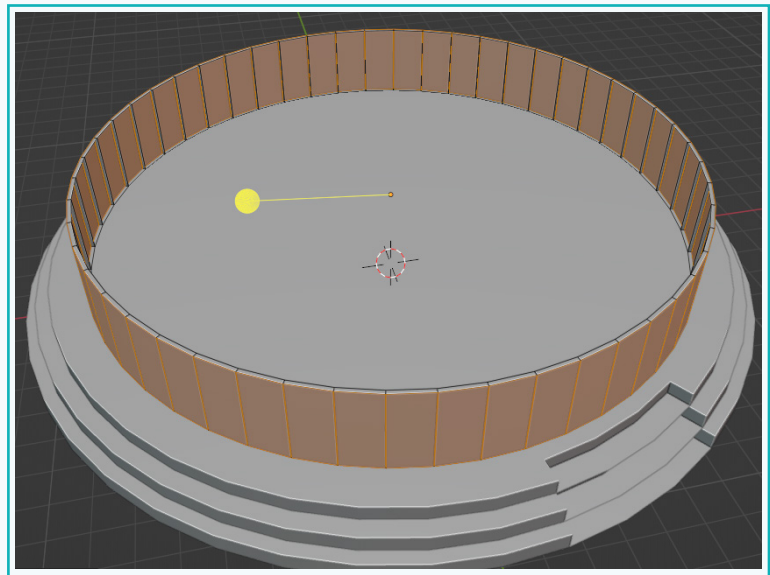
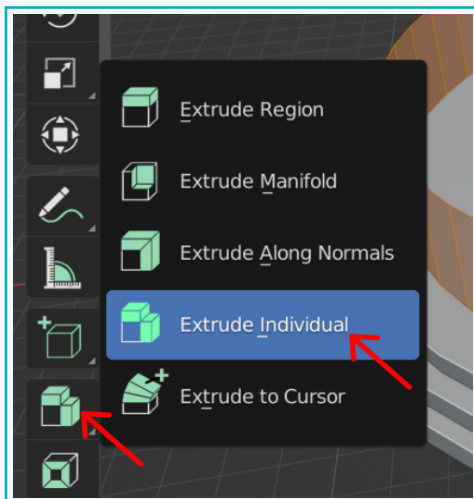


- 16 Next, select the face loop by double-clicking on the intersection between any two faces on the loop.

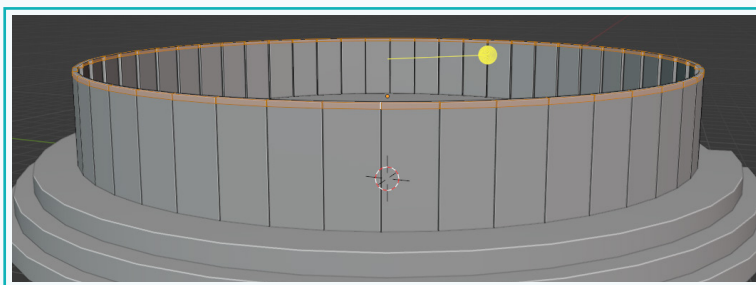
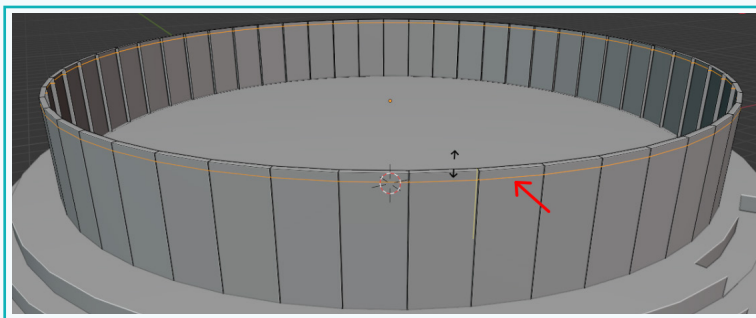


Activity

- 17 Next, long press on the 'Extrude' tool to get additional extrude options. Select 'Extrude Individual'. Extrude the selected face loop outwards a little bit.

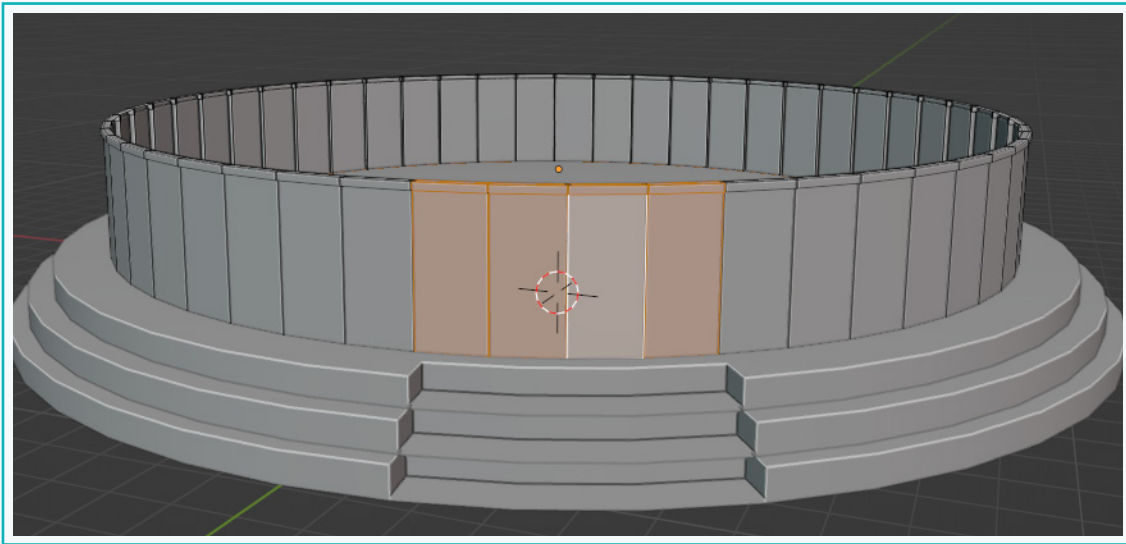


- 18 To add additional details, press 'Ctrl + R' to add edge loop, as shown. As done in the previous step, select the face loop and then extrude the individual faces outwards.

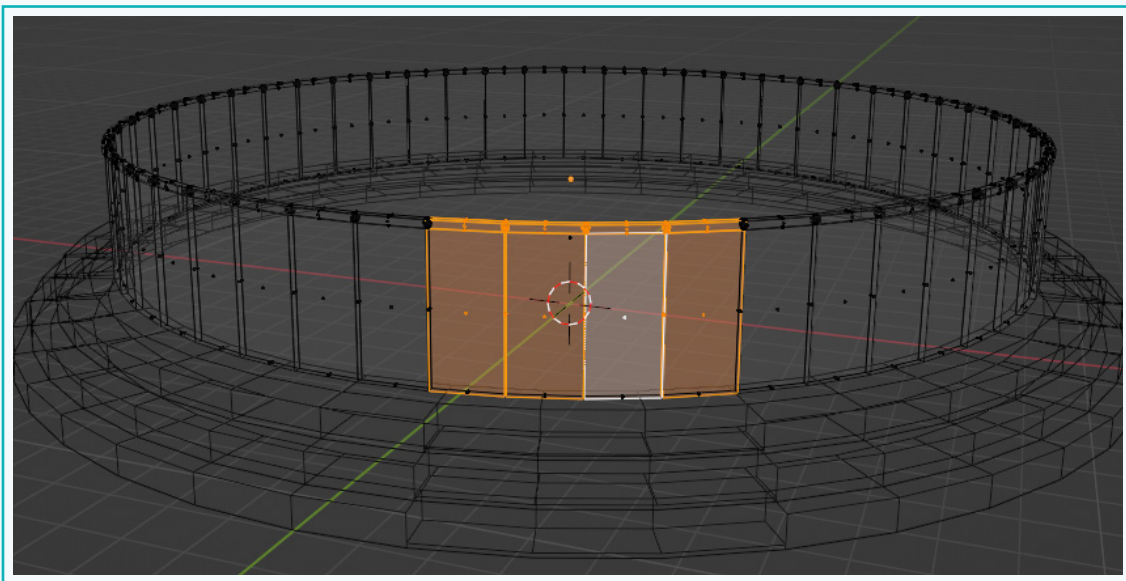


Activity

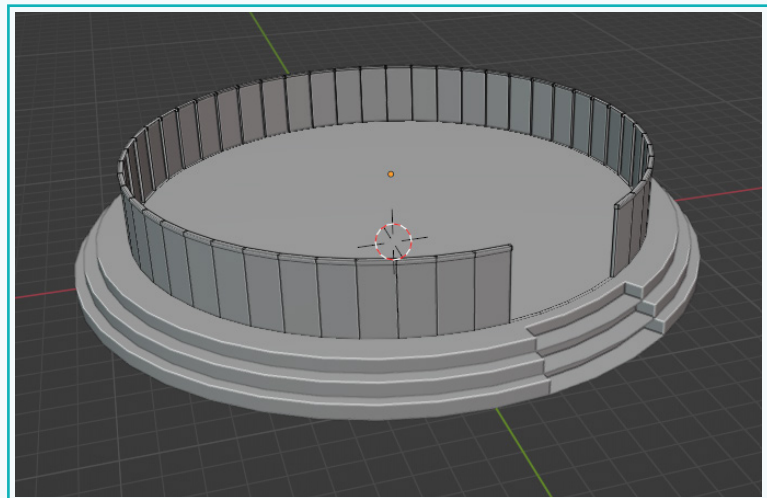
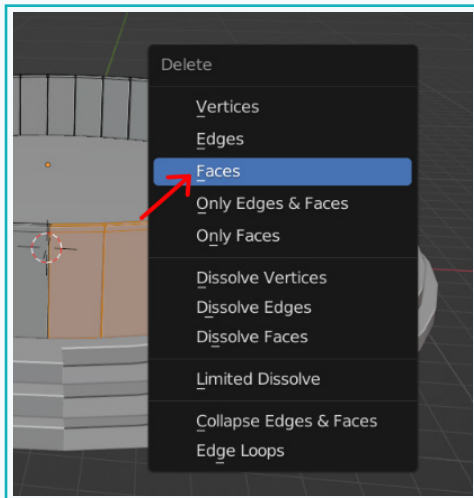
- 19 Now, let us create an opening for the entrance at the position where the stairs have been created. Select the faces corresponding to the steps.



- 20 To make sure that all the faces have been selected, go into the Wireframe mode, and use Box select option to select. Press 'X' to delete them.



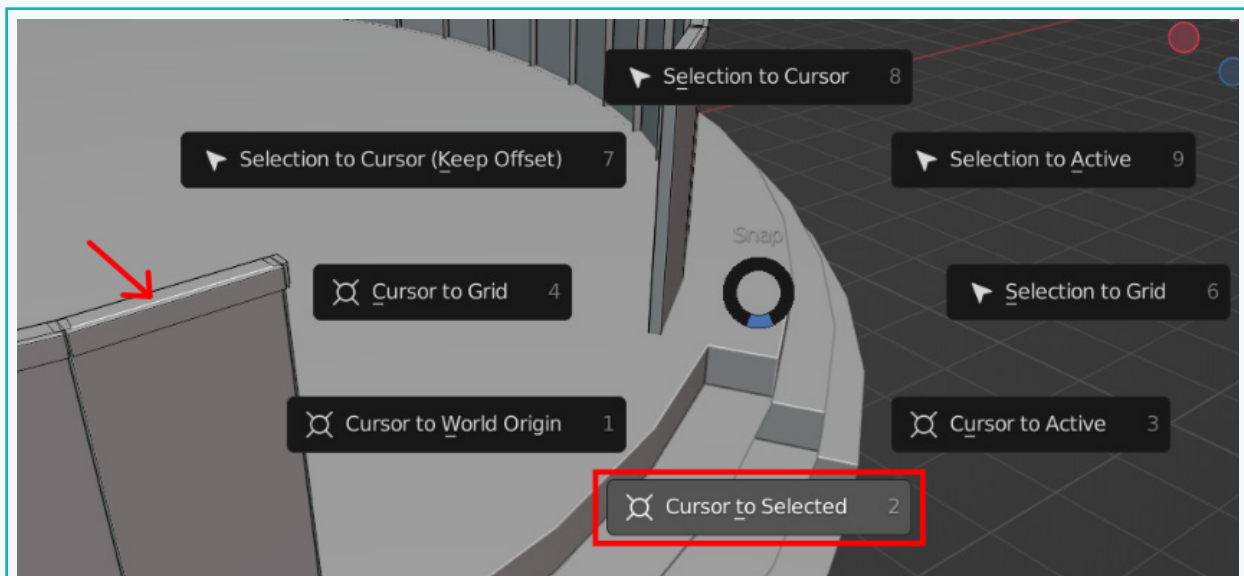
Activity



Next, we will create small arches encircling the entire perimeter of the cylindrical wall.

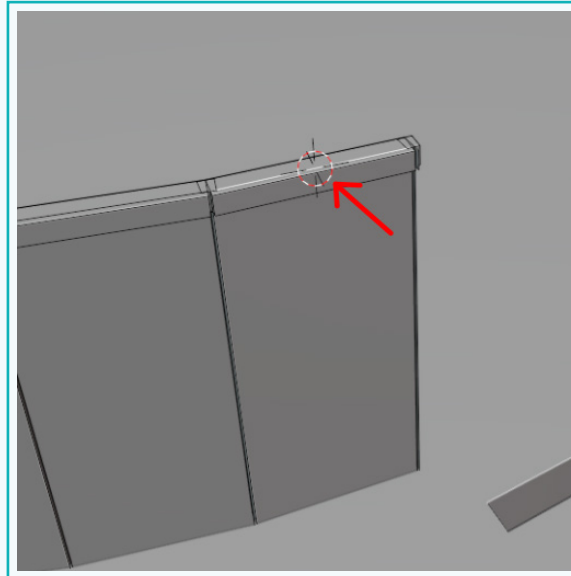
- 21** Select an edge closest to the entranceway, as shown. Press 'Shift + S' for the Snap menu and click on 'Cursor to Selected'.

Blender provides a dedicated snap menu for the 3D cursor. When used in conjunction with the snapping options, the 3D cursor can be used to align objects to specific locations in the scene.

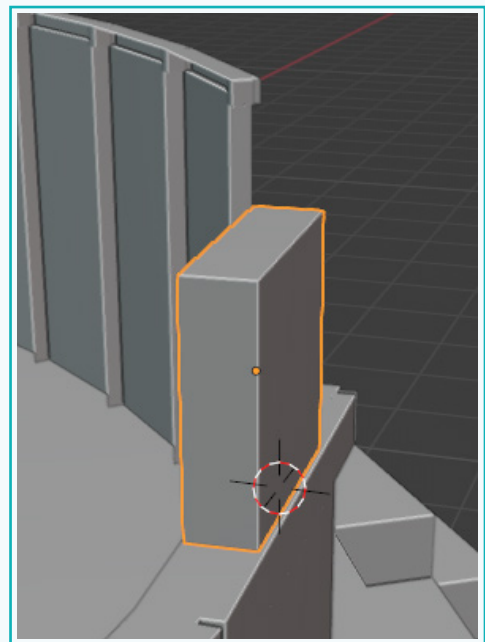
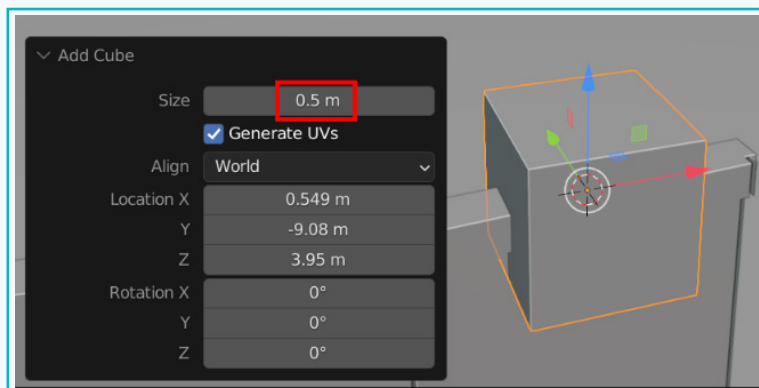


Activity

- This snaps the 3D cursor to the selected edge.

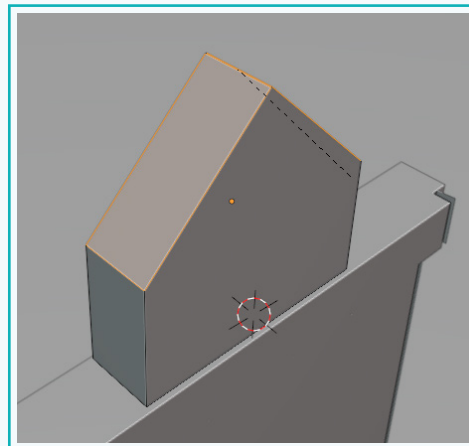
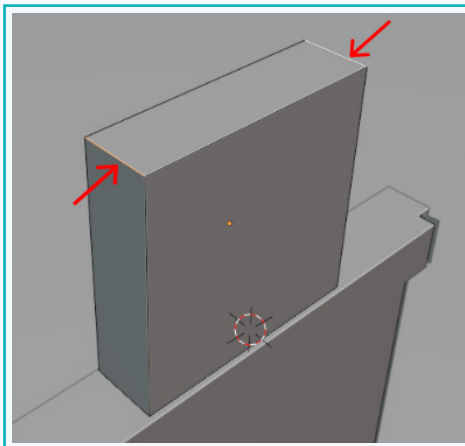


- 22** Create a cube with the specified dimensions. The cube gets created at the position of the 3D cursor. Position the cube on the wall and scale it on the y-axis to match the thickness of the wall.



Activity

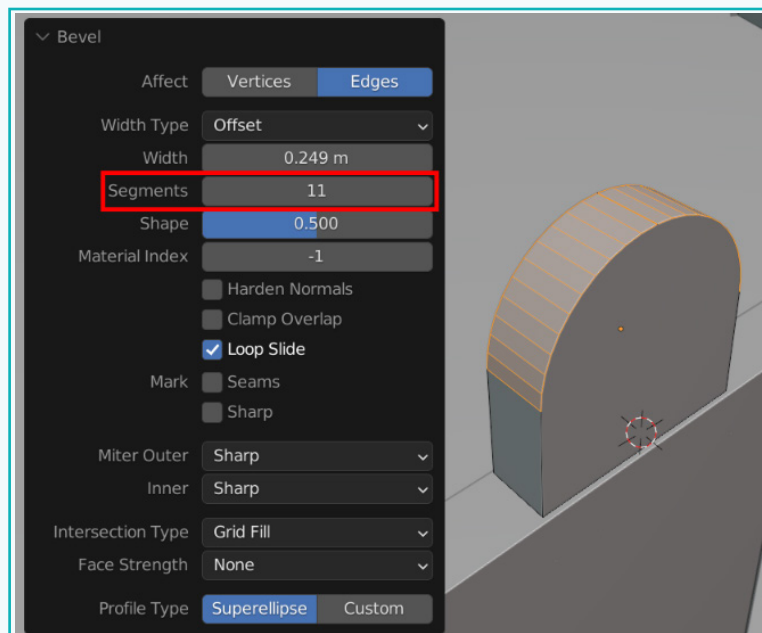
- 23 Press 'Tab' to go into the 'Edit' mode. Then, select the two edges shown in the image. Press 'Ctrl + B' to bevel the selected edges. Move the mouse to create the shape, as shown.



Bevel Tool

The 'Bevel' tool allows you to create rounded corners on geometry. A **bevel** is an effect that smooths out edges and corners to give a more refined and polished look.

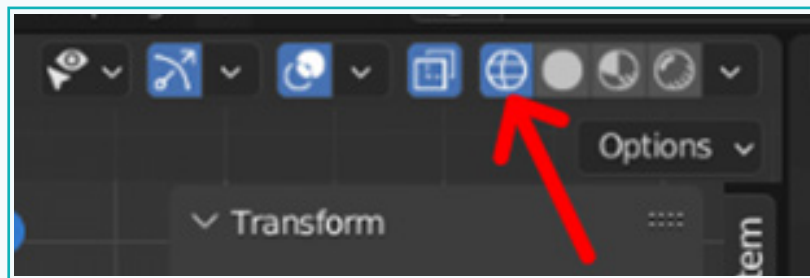
- 24 When you release the mouse, the 'Bevel' options will appear. Increase the value of 'Segments' to '11' to create a rounded bevel.



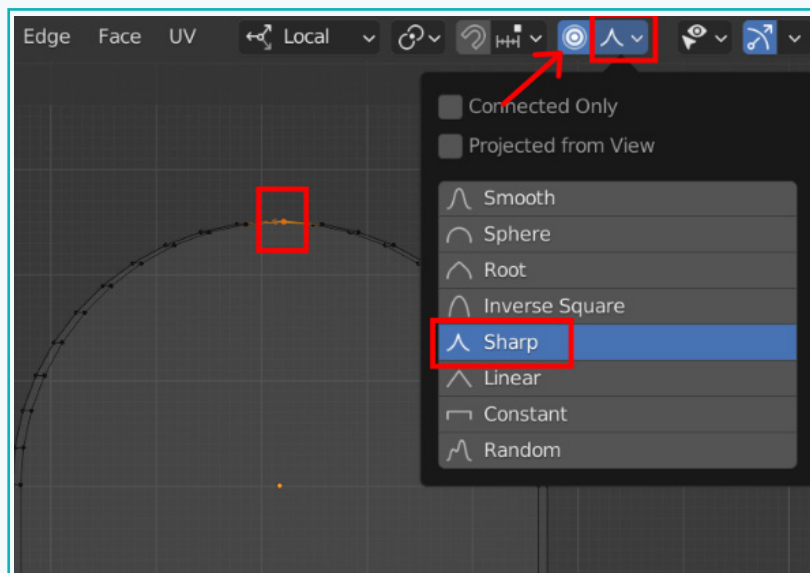
Activity

- 25 Next, we want to create a pointed top for the arch. For this, go into the front view and select the 'Wireframe Shading' mode.

The **Wireframe Shading** mode helps to get an x-ray view of the geometry and helps to view and select even those vertices/edges/faces which are not visible from the current view.



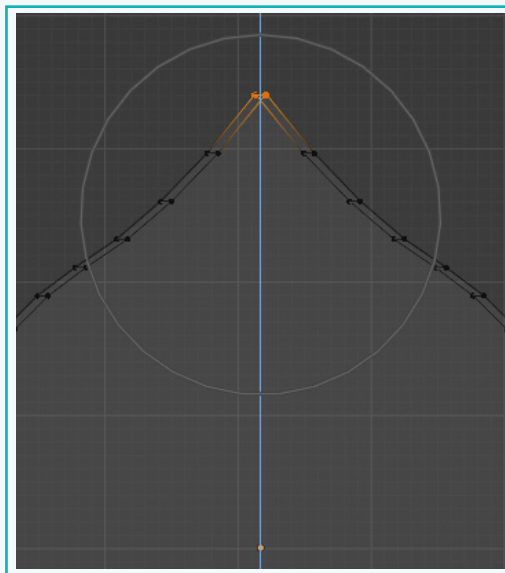
- Then, toggle on the 'Proportionate Editing' mode and select 'Sharp' from the dropdown options. Select the centre vertices, as shown.



- 26 Now, press 'G', followed by 'Z', to move the selected vertices upwards. Scrolling the mouse wheel helps to adjust the area of influence of the move operation.

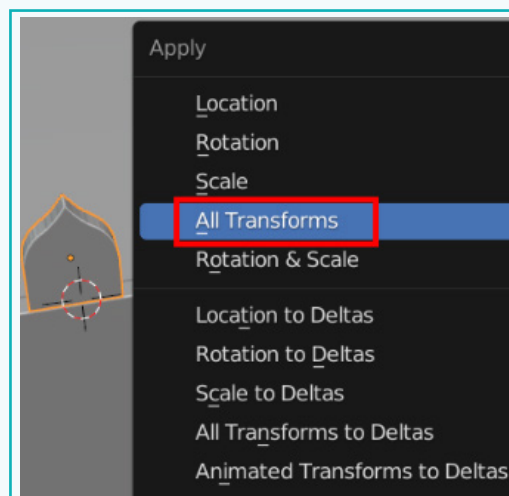
Activity

Proportional Editing is a technique that enables the manipulation of chosen elements, such as vertices, while having an impact on adjacent elements. This means that altering the position of one vertex within a certain range can also result in the movement of unselected vertices. This is particularly useful when we need to smoothly deform the surface of a mesh.



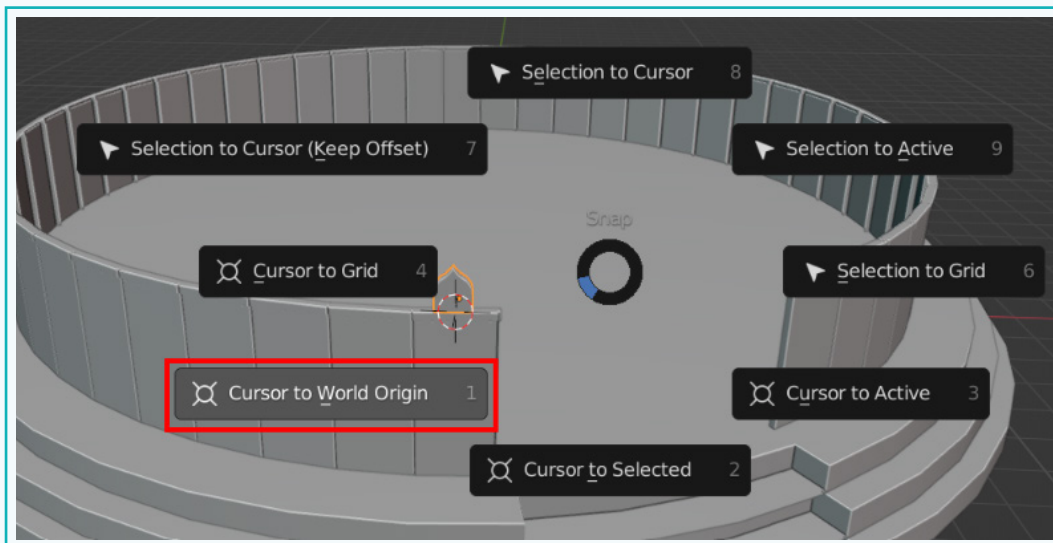
- 27 Once we get the desired shape of the arch, go into Object mode and press 'Ctrl + A' for the 'Apply' menu. Select 'All Transforms'.

This is an essential step and skipping it may result in the next few steps not working accurately. 'Applying Transforms' is essential for some of Blender's tools and modifiers to work accurately.

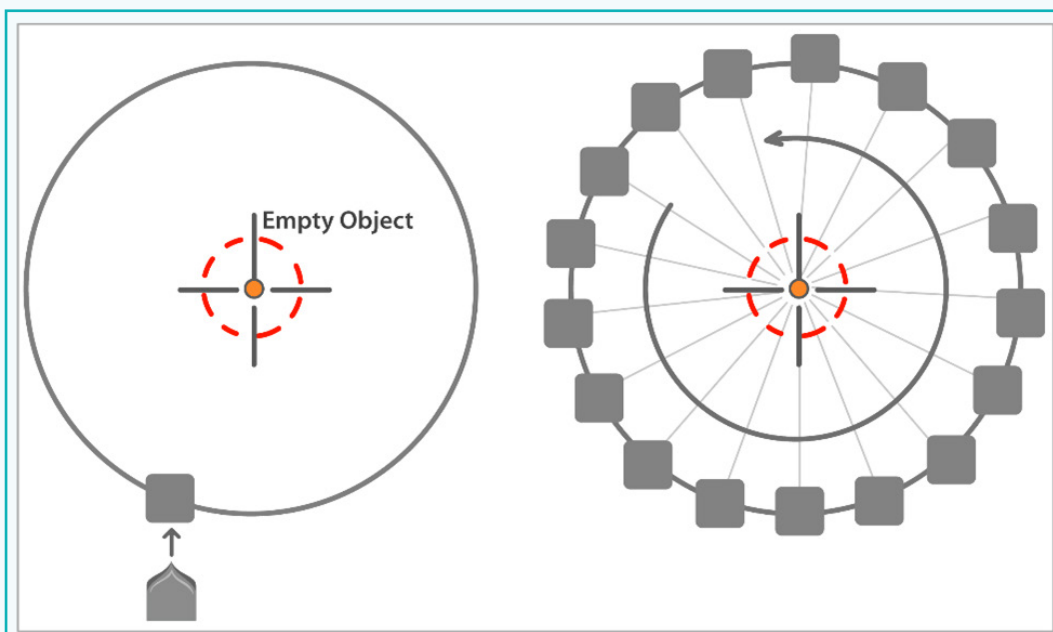


Activity

- 28 Press 'Shift + S' for the 'Snap' menu and select 'Cursor to World Origin'. The 3D cursor snaps onto the world origin.

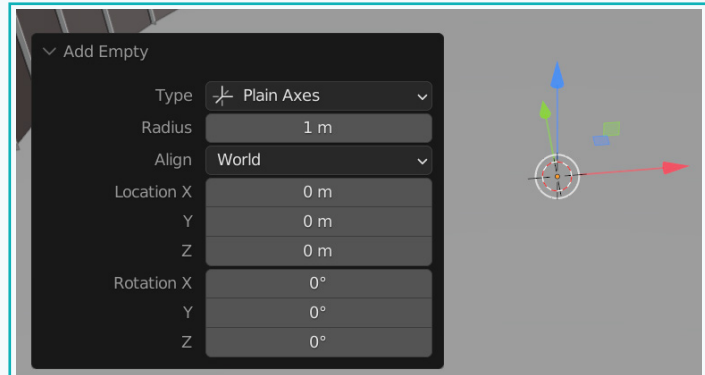
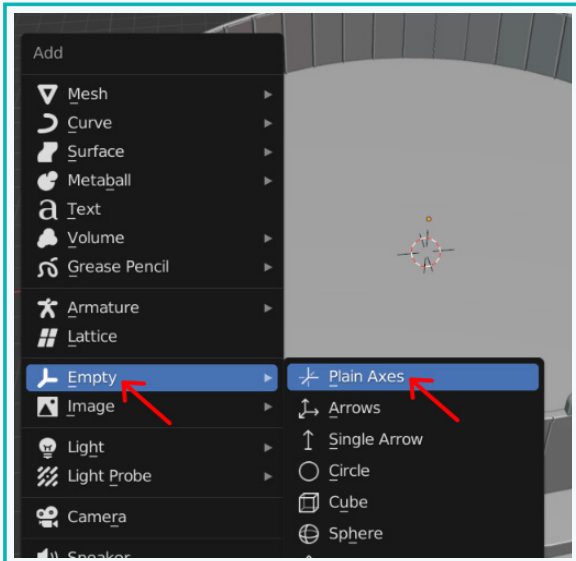


Next, we will create an empty object so that we can create an array of the arch with the empty object as the origin point.

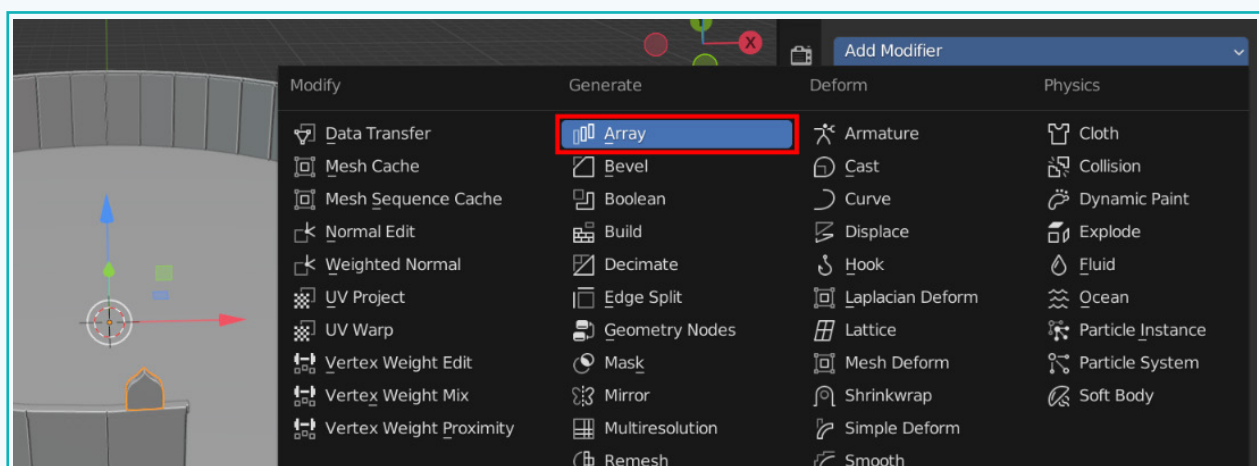


Activity

- 29 Press 'Shift + A' → 'Empty' → 'Plain Axes'. It gets created at the position of the 3D cursor.

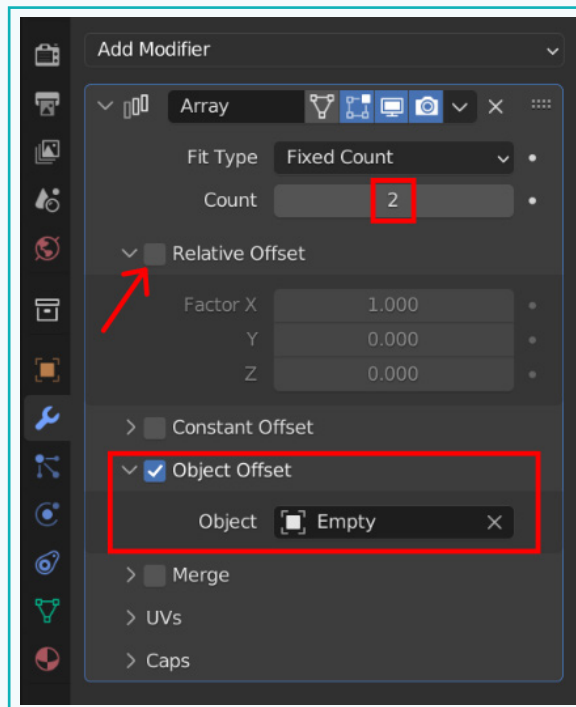


- 30 Next, select the arch, and click on the 'Modifiers' window to add a modifier. Select the 'Array' modifier.

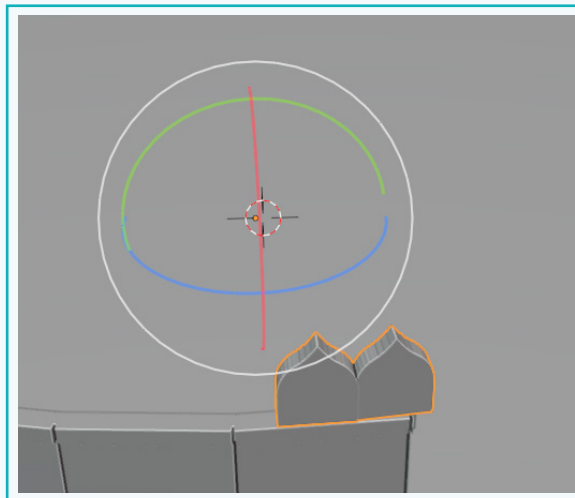


Activity

- 31** Uncheck the 'Relative Offset' option and enable the 'Object Offset' option. Under the 'Object' property, select the 'Empty' Plain Axes object that we created earlier. Note that the 'Count' value is '2'. I.e., an array of 2 arches will get created.

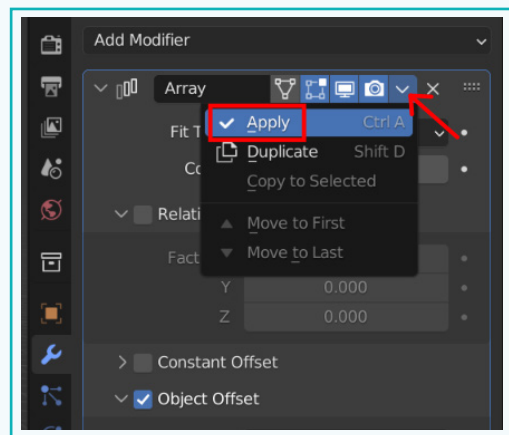
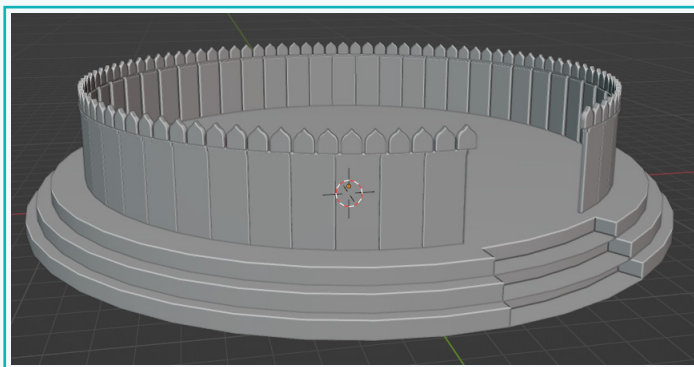
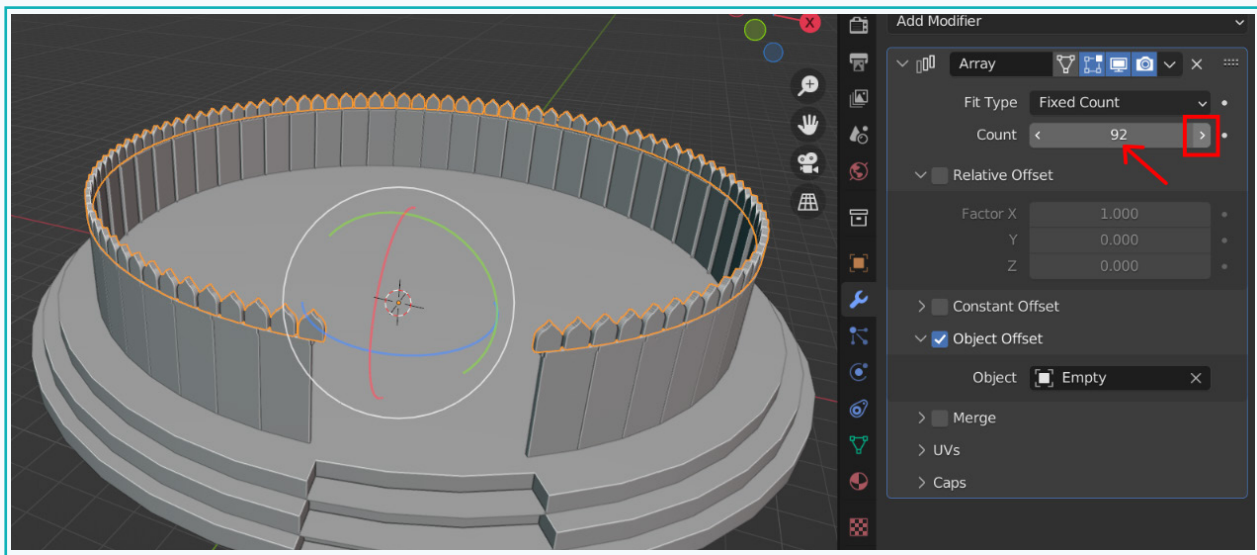


- 32** Now, select the arch and rotate it. The second arch will be visible.



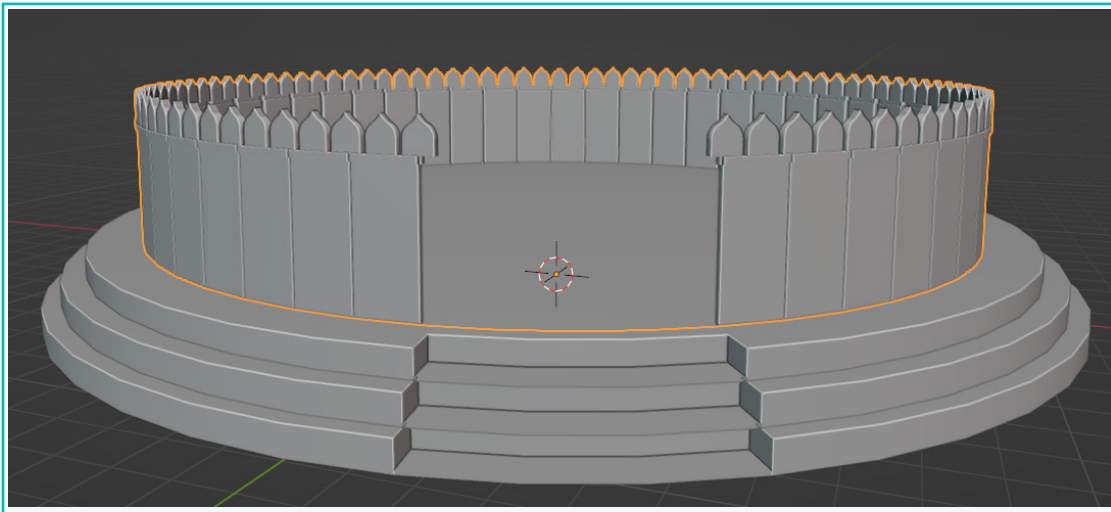
Activity

- 33 Now, we must keep increasing this 'Count' value until arches are constructed around the whole perimeter of the wall. Once done, apply the 'Array' modifier.

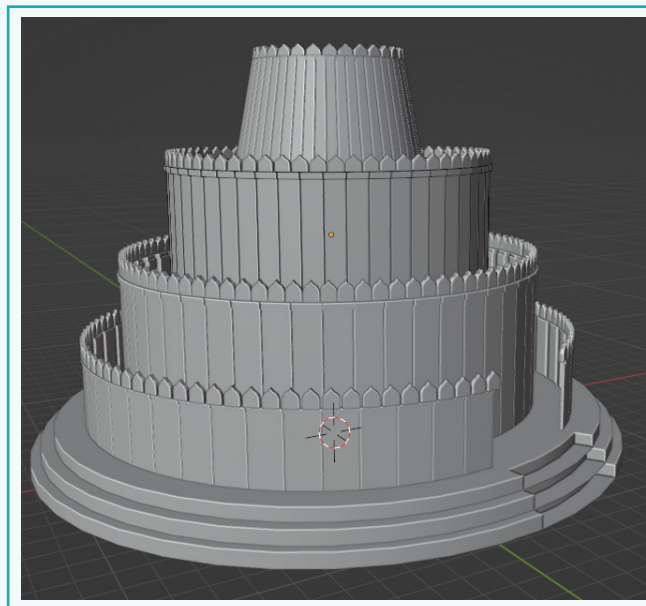


Activity

- 34** Next, select the cylindrical wall and the arches and press 'Ctrl + J' to join them and turn them into one single mesh.



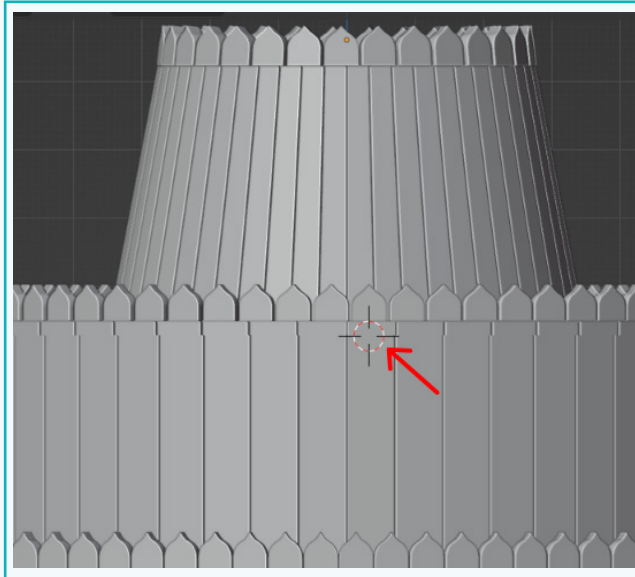
- 35** Repeat the same steps that we followed for this first level of the castle and create the remaining three levels of the fort.



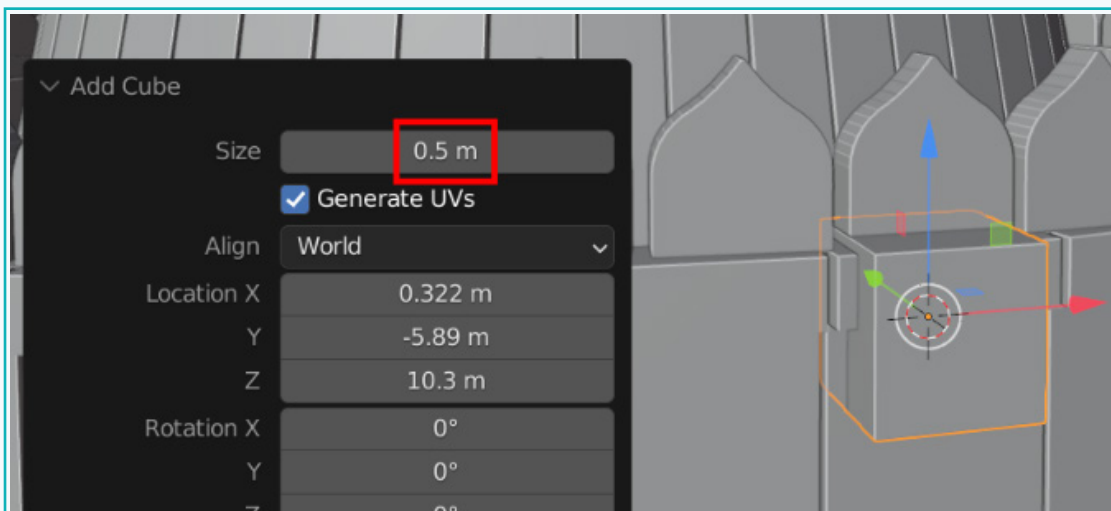
Next, let us create the arch-shaped openings on the top two levels of the castle.

Activity

- 36 Go into the front view and position the 3D cursor on the wall, as shown.

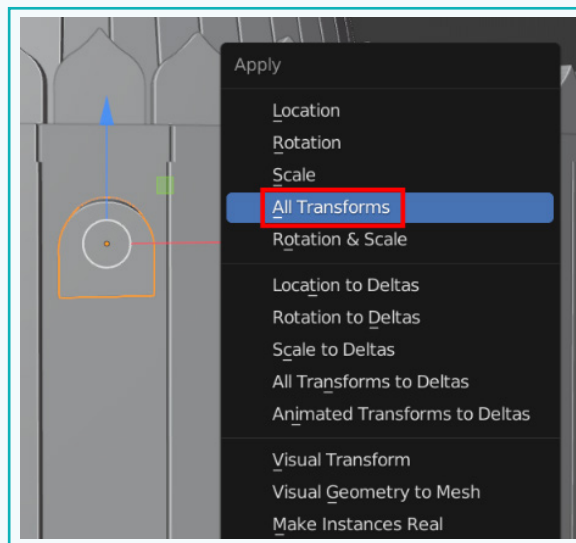
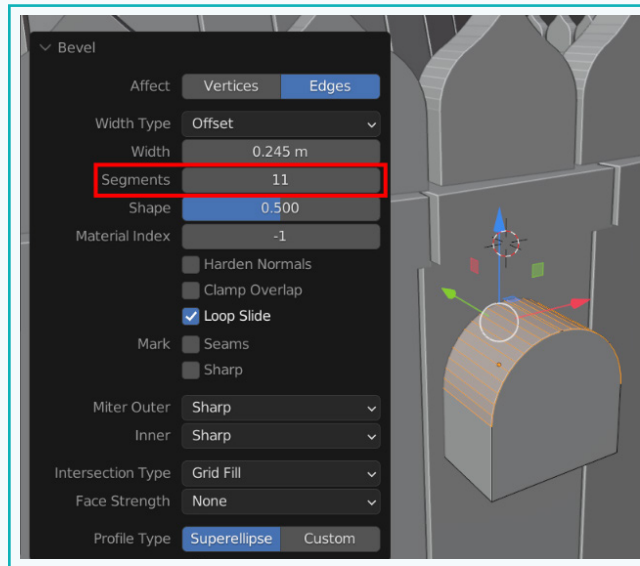
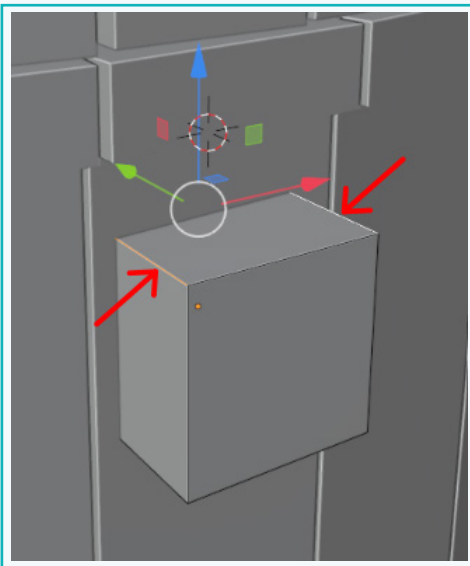


- 37 Create a cube with the specified dimensions. The cube gets created at the position of the 3D cursor.



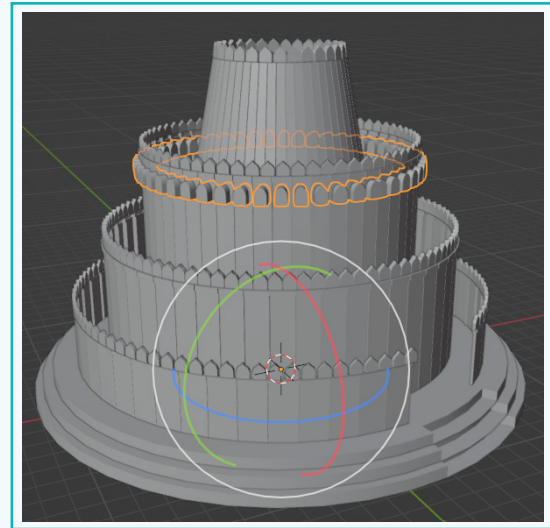
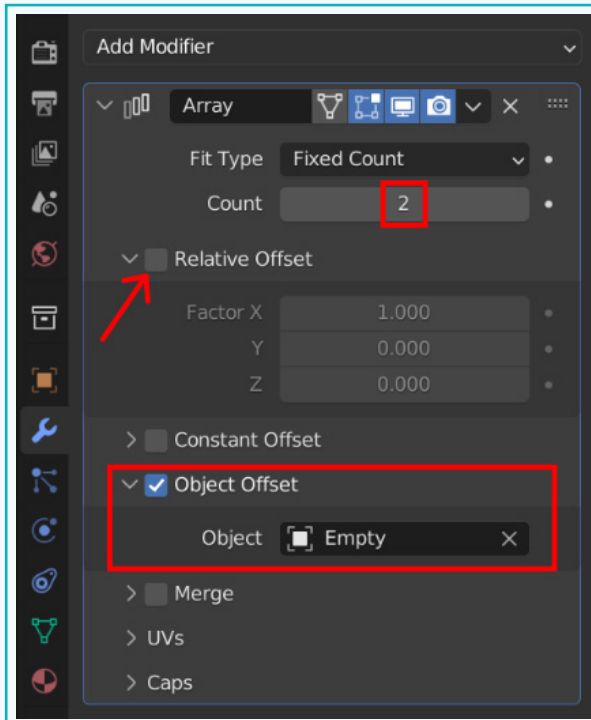
Activity

- 38 Press 'Tab' to go into the 'Edit' mode and select the two edges, as shown. Press 'Ctrl + B' to bevel the selected edges. Specify the 'Segments' value as '11'. Then go into the 'Object' mode and press 'Ctrl + A' and apply 'Transforms'.

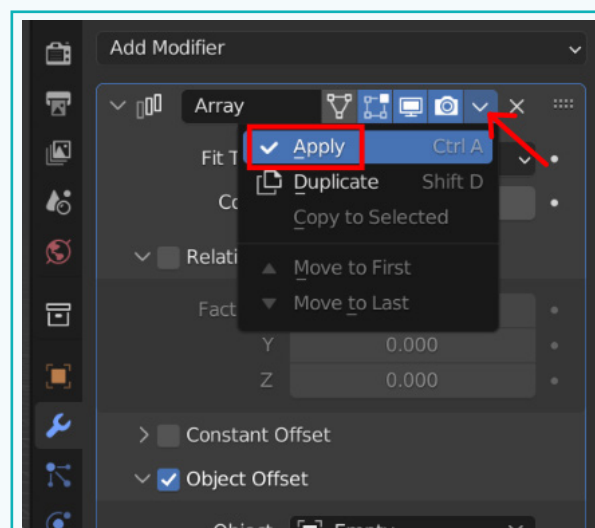


Activity

- 39 Create an array of the arch in the same way as you did before by using 'Object Offset'.

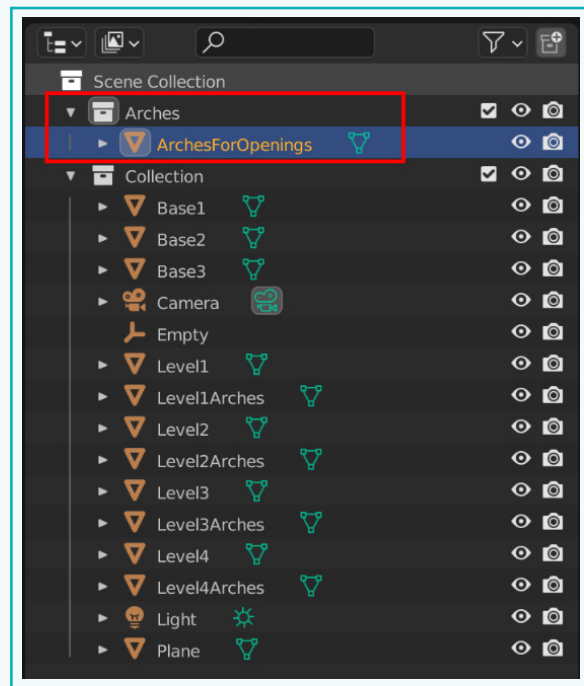
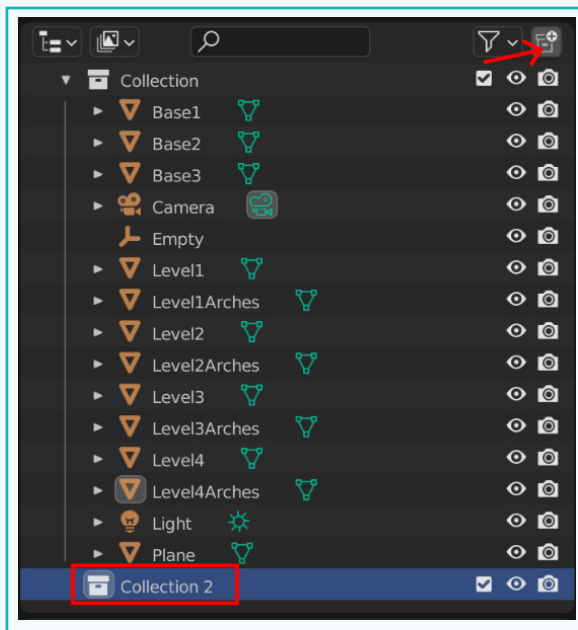


- 40 Apply the 'Array' modifier.

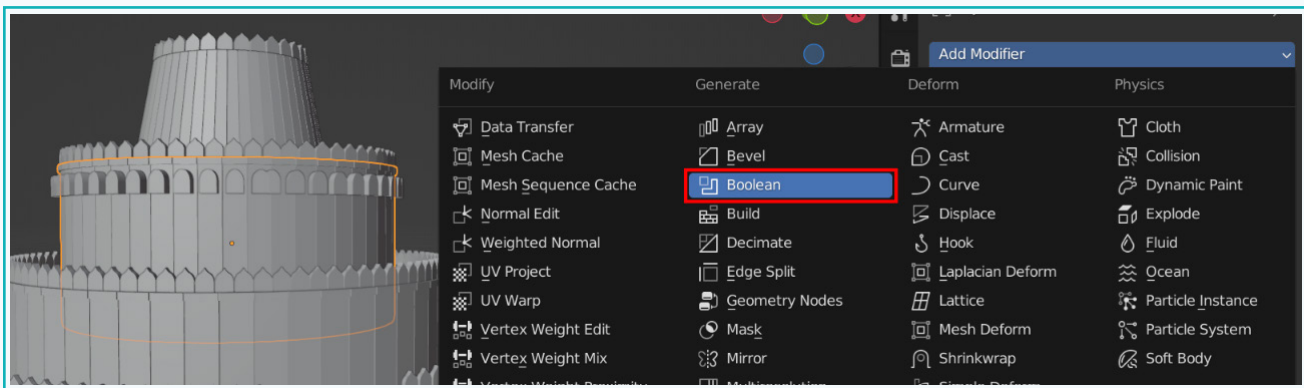


Activity

- 41 Next, create a new collection. Rename it as “Arches” and move the array of arches, created in the previous step, into it.



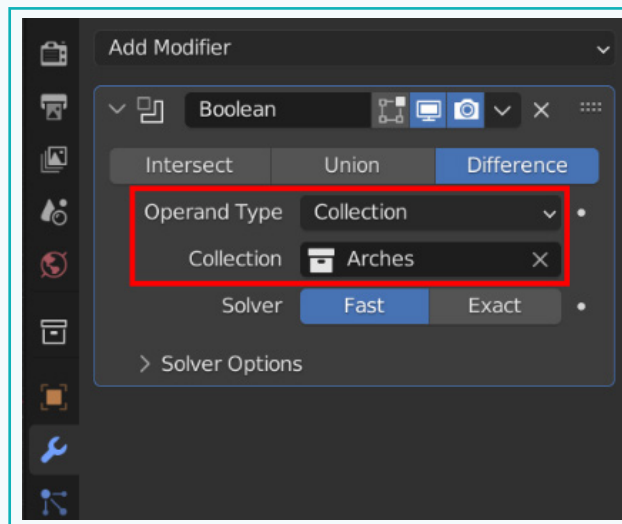
- 42 Next with the level 3 cylinder selected, add the ‘Boolean’ modifier to it.



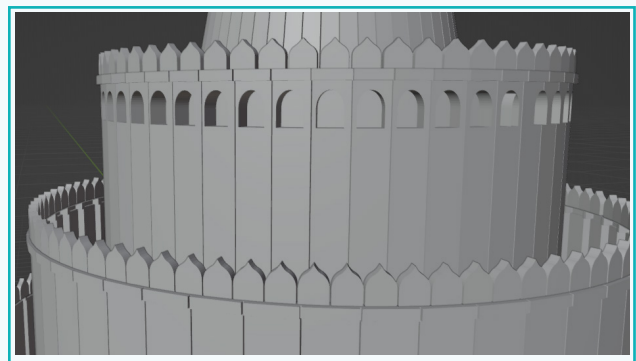
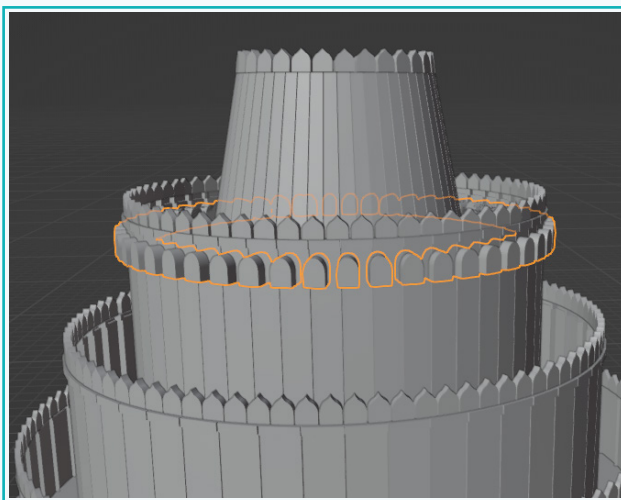
A Boolean modifier is a powerful tool that enables us to execute Boolean operations on our 3D geometry. By utilizing this tool, we can combine two intersecting objects to create a Boolean, and the result depends on the operation applied. Different Boolean operations produce distinct results, offering a range of possibilities for manipulating our 3D models.

Activity

- 43 Set the 'Operand Type' to 'Collection' and set the 'Collection' to 'Arches'. Make sure the 'Difference' tab is selected at the top and the 'Solver' option is set to 'Fast'.

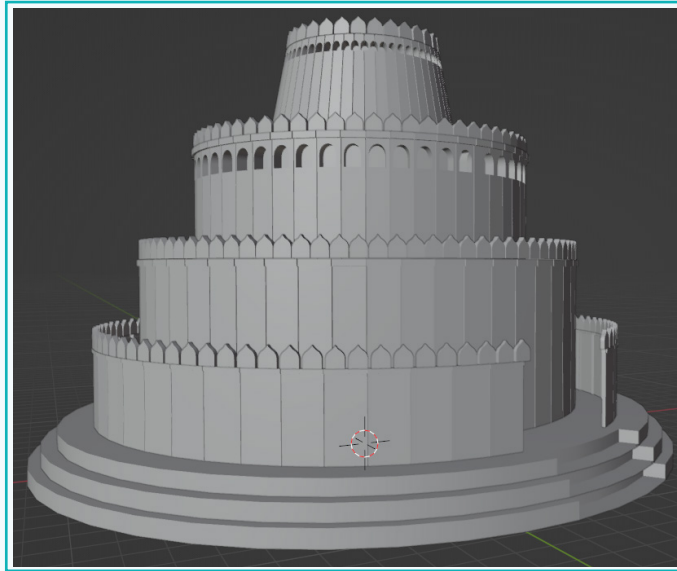


- 44 The shape of the array of arches has been subtracted from the shape of the cylindrical wall. To see the effect of the modifier, select the array of arches and press 'H' to hide it.



Activity

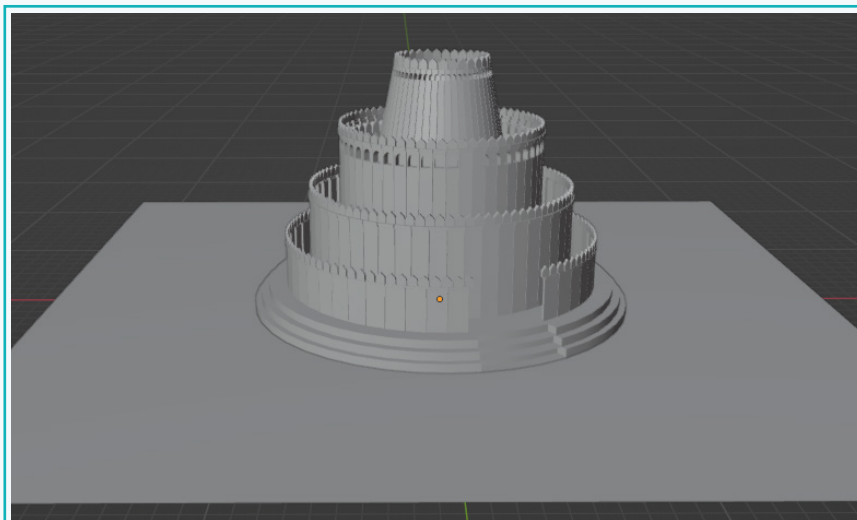
- 45 Repeat the same steps to create the arched openings for the topmost level of the structure.



Create the Environment

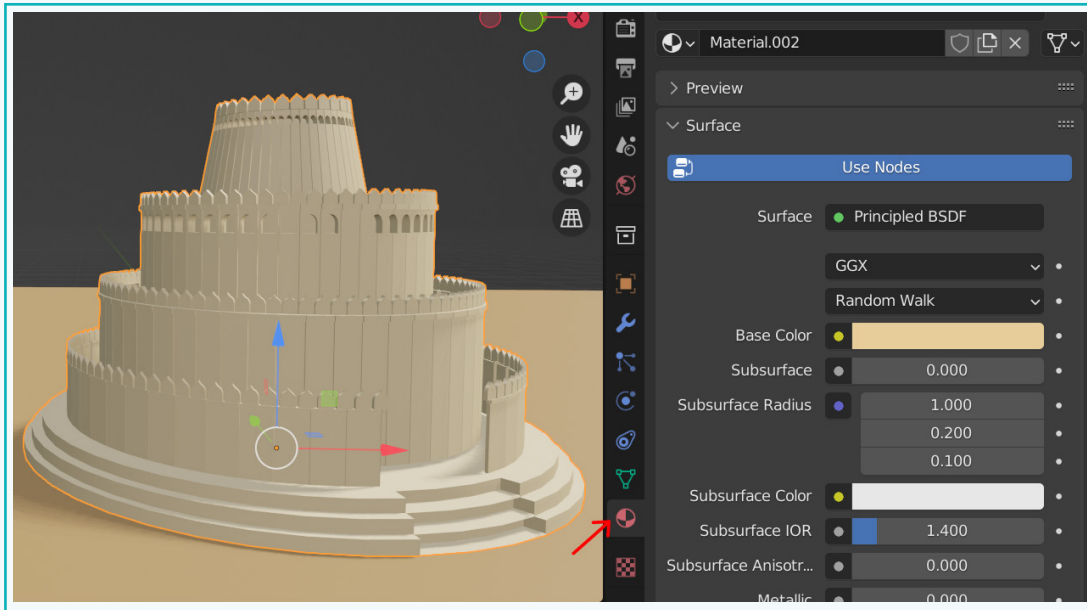
Next, let us create an environment for the architectural structure so that we can then render the scene.

- 1 First, let us create a plane to represent the ground.



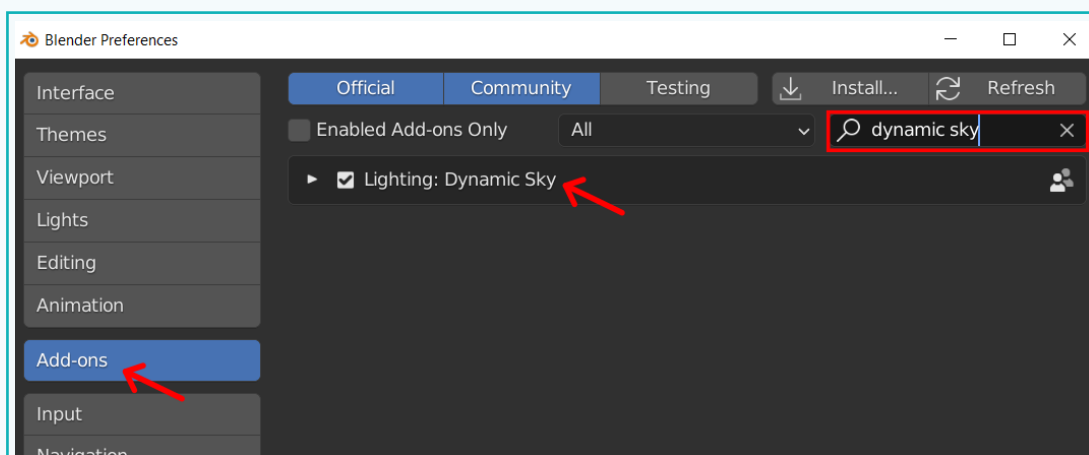
Activity

- 2 Next, add materials to the ground and the structure. Select a suitable colour.



Finally, to make it look realistic, let us create a sky background for our scene by making use of an add-on provided by Blender – Dynamic Sky.

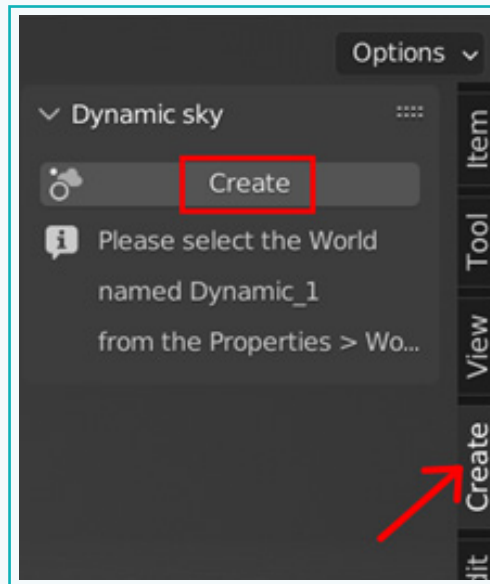
- 3 Go to 'Edit' → 'Preferences' → 'Add-ons'. Type 'dynamic sky' in the search box and enable the add-on from the list below.



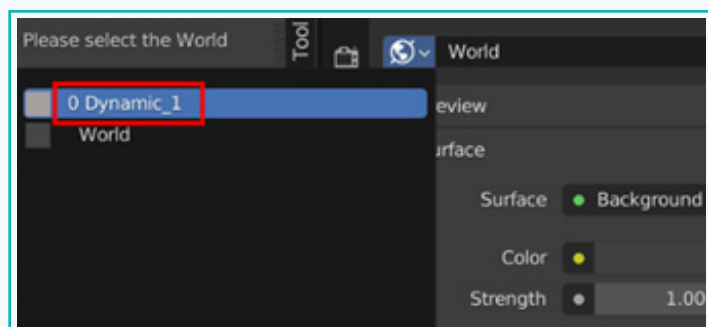
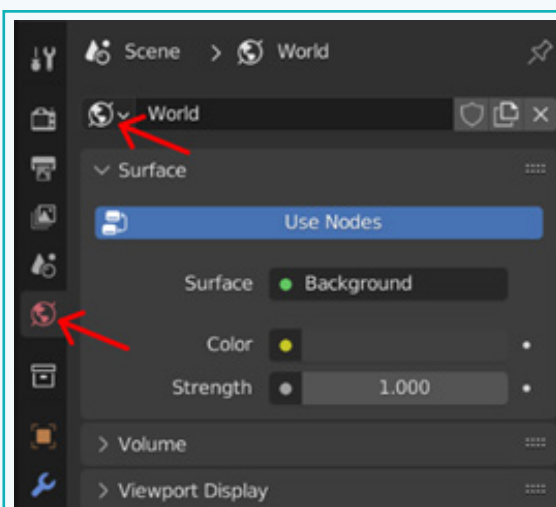
Once enabled, a new tab called 'Create' appears in the side panel area of the 3D workspace.

Activity

- 4 Click on the 'Create' tab. Then, click on the 'Create' button under the 'Dynamic Sky' dropdown.



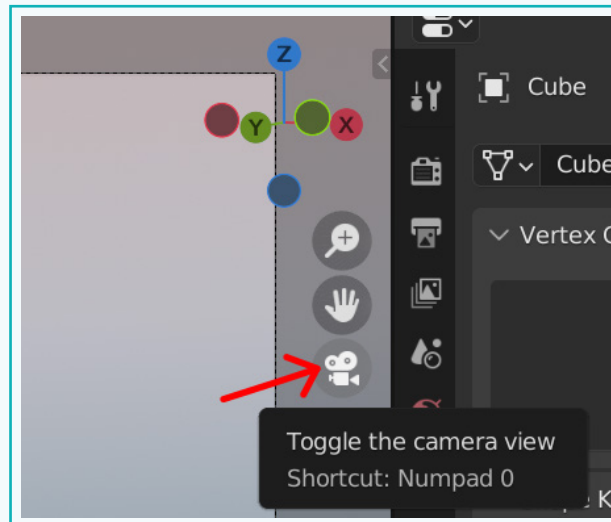
- 5 Next, go into the 'World' properties window and click on the dropdown next to the name of the world. Select 'Dynamic_1' from the list.



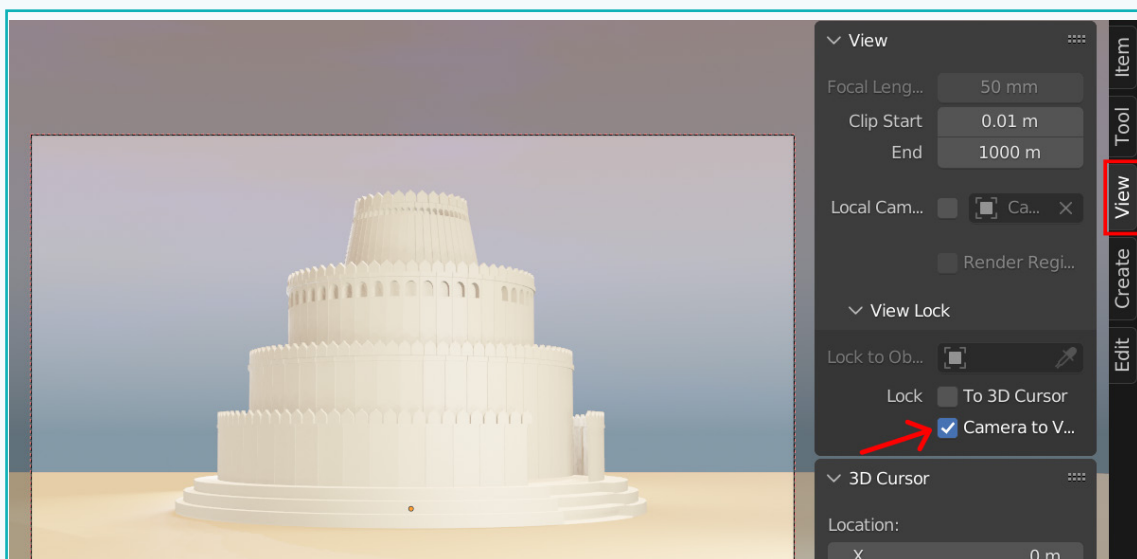
To be able to render the scene out, our object of interest should be positioned accurately within the camera view.

Activity

- 6 Click on the 'Camera' icon in the 3D viewport. The camera view will be displayed.



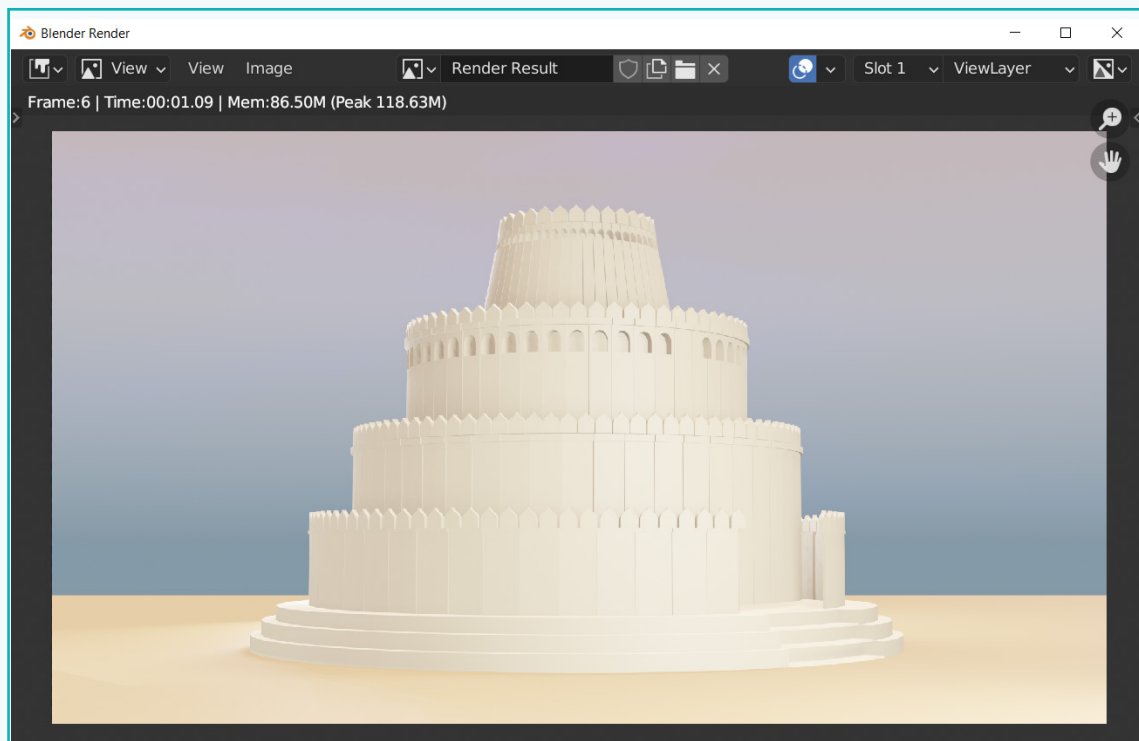
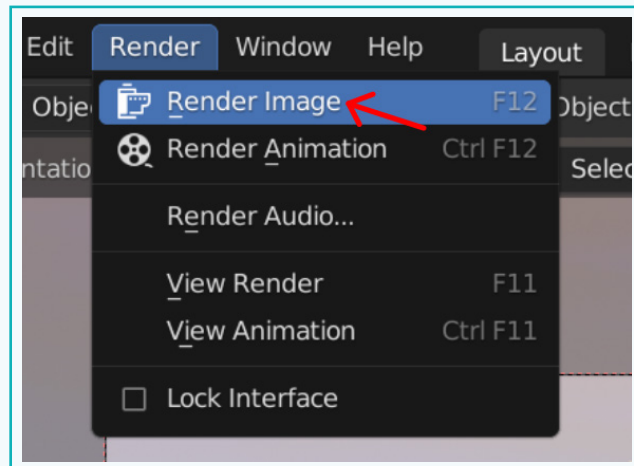
- 7 To change the view within the camera and position our 3D model well within the camera view, click on the 'View' tab on the side panel and enable the 'Camera to View' option. This allows the camera to remain fixed while we reposition the view to fit within the camera view.



Finally, we can now render the scene out.

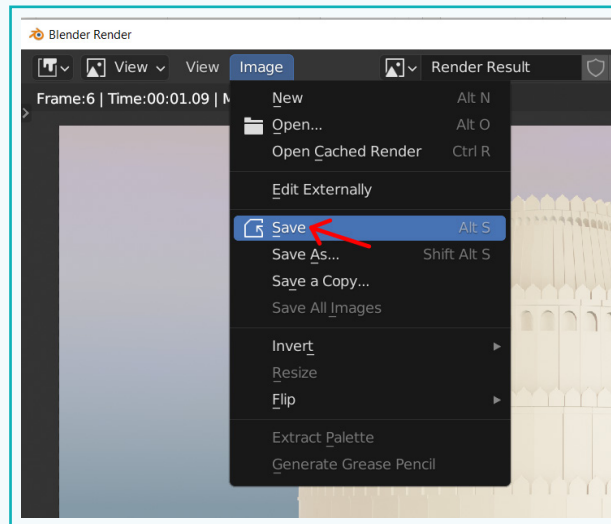
Activity

- 8 Click on the 'Render' menu → 'Render Image'. Our scene gets rendered.

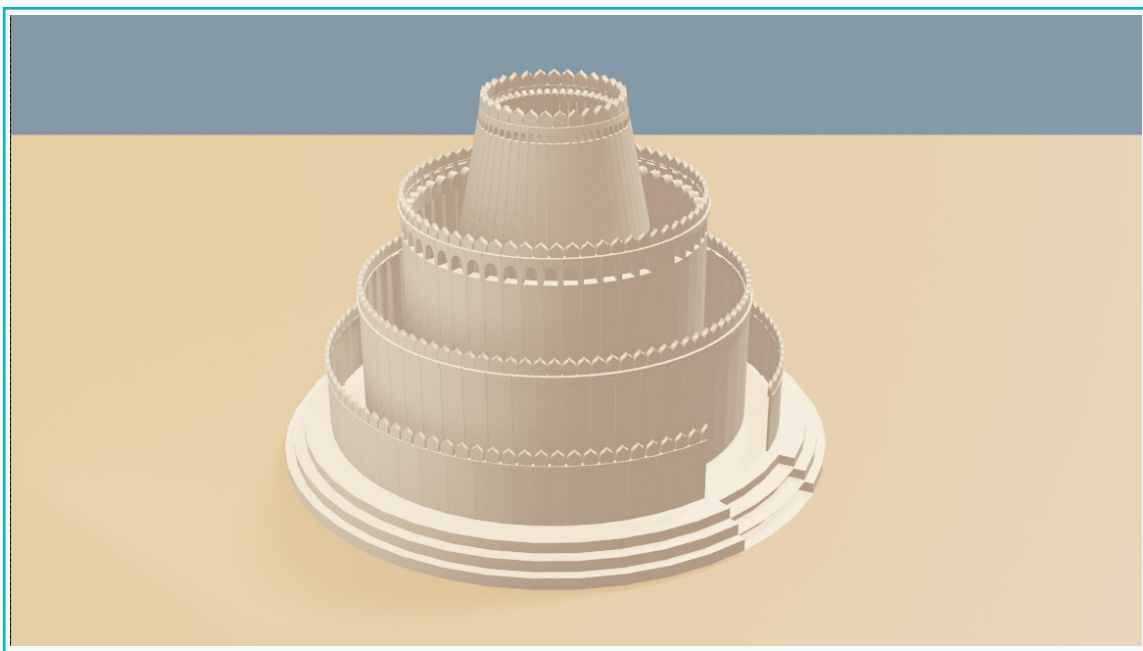


Activity

- 9 We can also save this image by clicking on 'Image' → 'Save'.



Final Output.



Exercise

Fill in the Blanks

- 1 The _____ tool creates a new face on the surface of the selected face.
- 2 The _____ tool is an edit mode tool meant for extending a selected portion of an object.
- 3 The _____ menu provides options to switch between different views.
- 4 The _____ tool allows you to create rounded corners on geometry.
- 5 The _____ Shading mode helps to get an x-ray view of the geometry.

Tick mark ☒ the Correct Answer

- 6 The shortcut for adding edge loops is -

a. Ctrl + R	☐	b. Alt + R	☐
c. Ctrl + E	☐	d. Alt + E	☐
- 7 The dedicated menu for the 3d cursor is called as the

a. Snap menu.	☐	b. Edit menu.	☐
c. Apply menu.	☐	d. Transform menu.	☐

Exercise

Provide One Line Answers

8 What is the Proportional Editing mode?



9 What is the Boolean modifier?



In this lesson, we explored how we can use Blender to create a 3D model of an architectural structure. We implemented interesting tools and modifiers within Blender like the Array modifier, Boolean Modifier, and other such techniques to create our 3D model.

In this way, creating a prototype and rendering it, gives us an idea of our architecture. We can make changes and try different looks as well.



Tech Flash

- **Inset** : The Inset tool creates a new face on the surface of the selected face.
- **Extrude** : The Extrude tool extends a selected portion of an object.
- **Wireframe** : The Wireframe Shading mode helps to get an x-ray view of the geometry.
- **Snap menu** : Snap menu is a menu dedicated for the 3D cursor.
- **Proportional Editing** : Proportional Editing is a technique that enables the manipulation of chosen elements while also affecting adjacent elements.
- **Boolean** : Boolean modifier is a powerful tool that enables us to execute Boolean operations on our 3D geometry.